

A Project Report

on

ORGAN-DONATION USING BLOCKCHAIN

Submitted to JNTUK – Kakinada

In the partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

Computer Science and Engineering (CSE)

by

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The project embodies results of original work, and studies are carried out by the students them self and the contents of this project work do not submit for the award of any other degree to this University/Institution or any other University/Institution.

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ACKNOWLEDGEMENT

We sincerely thank the following distinguished personalities who have given their advice and support for successful completion of the work.

We wish to express our deep sense of gratitude by our guide [Mr.D.V.Srinivasa Rao,M.Tech,Associate Professor](#) , Department of C.S.E, NARASARAOPETA INSTITUTE OF TECHNOLOGY, for giving us an opportunity to work under her by giving valuable suggestions and kind cooperation, guidance throughout the time. Her vast experience, profound knowledge and willingness have been a constant source of inspiration for us throughout this project work. My sincere thanks to him for evincing keen interest in our endeavour and for being very amicable.

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DECLARATION

This is to certify that the work reported in project titled, entitled “**ORGAN-DONATION USING BLOCKCHAIN**” submitted in the Department of Computer Science and Engineering (CSE), **NARASARAOPETA INSTITUTE OF TECHNOLOGY, NARASARAOPET**, in the partial fulfillment of degree for the award of Bachelor of Technology, and is a bonafied work done by us.

These reported results are based on the project work entirely done by us and not copied from any other source. Also, we declare that the matter embedded in this project has not been submitted by us in full or partial thereof for the award of any degree/diploma of any other institution or University previously.

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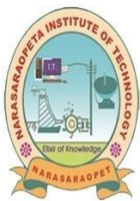
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PSO4	Conduct Investigation of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
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PSO6	The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PSO7	Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of and need for sustainable development.
PSO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PSO9	Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PSO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PSO11	Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
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C423.2	Perform requirement analysis and identify design methodologies	Analyze-L3
C423.3	Develop the product by using Hardware and Software tools	Analyze-L6
C423.4	Prepare Technical report about the project	Evaluate-L5

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C423.2	3	3	3	2	2	-	-	-	-	-	-	-	3	3
C423.3	2	4	3	2	3	-	-	-	-	-	-	-	3	3
C423.4	2	3	2	2	2	-	-	-	-	-	-	-	3	3
C423.5	3	3	2	2	2	-	-	-	-	-	-	-	3	3

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ABSTRACT

The process of organ donation faces several challenges, including issues of transparency, data security, and trust among various stakeholders such as donors, recipients, hospitals, and regulatory bodies. To address these issues, blockchain technology offers a promising solution by providing a secure, transparent, and decentralized platform for managing organ donation records. This paper proposes a blockchain-based system for organ donation, aiming to improve the efficiency, transparency, and trustworthiness of the organ donation process. By using blockchain's immutable ledger, the system ensures that donor information, organ availability, and recipient eligibility are securely recorded and shared in a tamper-proof manner.

Additionally, smart contracts can automate and enforce ethical rules and regulations related to organ donation, minimizing human errors and biases in decision-making. The system also enables real-time tracking of organ transfers, ensuring that organs are allocated in the most efficient and equitable manner. Through decentralization, blockchain can provide a trusted framework that reduces fraud, enhances privacy protection, and promotes accountability, thereby improving the overall transparency of the organ donation system. This approach has the potential to streamline the organ donation process, reduce waiting times for recipients, and ultimately save lives.

Blockchain provides a decentralized and immutable ledger that ensures all donor and recipient data is securely stored and cannot be altered or misused. In this system, donors can register their details, and hospitals can update patient requirements in real time. Smart contracts are used to automate the verification process and match donors with recipients based on medical compatibility, urgency, and location.

The proposed system enhances trust among stakeholders by ensuring transparency and traceability of every transaction in the organ donation process. It also reduces fraud, prevents unauthorized access, and improves the efficiency of organ allocation. By leveraging blockchain technology, this project aims to create a reliable, ethical, and secure platform that promotes organ donation and ultimately saves more lives.

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CHAPTER 1

INTRODUCTION

.INTRODUCTION:

Organ donation is one of the most important medical practices that helps save lives by transplanting healthy organs from donors to patients in need. However, the current organ donation system faces several challenges, including lack of transparency, data security issues, illegal organ trading, and delays in donor-recipient matching. These problems reduce public trust and limit the effectiveness of organ donation systems.

With the advancement of technology, blockchain has emerged as a powerful solution to address these issues. Blockchain is a decentralized and secure technology that stores data in a distributed ledger, ensuring that information cannot be altered or tampered with. This makes it highly suitable for sensitive applications like healthcare and organ donation management.

In this project, a blockchain-based system is proposed to improve the organ donation process. The system allows donors to register securely, hospitals to update patient data, and authorities to monitor the entire process transparently. Smart contracts are used to automate donor-recipient matching based on compatibility, urgency, and availability, reducing manual intervention and delays.

By implementing blockchain technology, the system aims to provide a transparent, secure, and efficient platform that enhances trust among users, prevents fraud, and ensures ethical organ allocation. This approach not only improves the overall efficiency of organ donation but also encourages more people to participate, ultimately helping save more lives.

Traditional organ donation systems are mostly centralized, where data is controlled by a single authority such as hospitals or government organizations. This centralization makes the system vulnerable to data breaches, manipulation, and lack of transparency. Issues such as illegal organ trafficking, unauthorized data access, fake registrations, and delays in verification processes further weaken the system. Additionally, the absence of a unified and secure platform for sharing donor and recipient information leads to inefficiencies in organ allocation.

CHAPTER 2

LITERATURE SURVEY

LITERATURE SURVEY :

1.Title: Blockchain for Secure and Transparent Organ Donation Systems

- **Authors:** M. E. S. Al-Bassam, A. I. Al-Kahtani, H. M. Al-Sahaf, et al. (2020)
- **Description:** This study explores how blockchain technology can enhance the transparency and security of organ donation processes. It discusses the implementation of decentralized records for tracking organ matches, donor consent, and patient information, ensuring that the entire process is secure, immutable, and auditable. The paper highlights the role of blockchain in reducing fraud and inefficiencies in the allocation of organs.

2.Title: A Blockchain-Based Framework for Organ Donation and Transplantation

- **Authors:** S. H. R. Reza, H. A. Kiani, A. V. Dastjerdi (2021)
- **Description:** This paper presents a framework for using blockchain to streamline organ donation and transplantation processes. It focuses on how smart contracts can automate and verify various stages of organ donation, from donor registration and consent to organ matching and allocation. The study aims to reduce human error, improve operational efficiency, and ensure data integrity.

3.Title: Blockchain in Healthcare: A Systematic Review and Application to Organ Donation

- **Authors:** S. S. Jain, P. L. Gohel, K. B. Pradhan (2020)
- **Description:** This review paper systematically analyzes the applications of blockchain in healthcare and specifically in organ donation systems. It provides an overview of the challenges faced by current organ donation frameworks, such as inefficiencies and data security issues, and explains how blockchain's decentralized structure can address these challenges. The paper discusses use cases, benefits, and potential limitations of blockchain in organ donation.

4.Title: Blockchain-Based Decentralized Solution for Organ Transplantation Transparency and Security

- **Authors:** T. S. Hussein, L. F. Souza, A. R. Fernandes (2021)
- **Description:** This research focuses on creating a decentralized system for organ transplantation using blockchain to increase transparency and security in organ donation. It highlights the potential of using blockchain to track the entire donation process—from initial consent to the final transplantation—ensuring that the process remains tamper-proof and transparent. The authors emphasize blockchain’s ability to build trust in the organ donation system.

5.Title: Smart Contracts and Blockchain in Organ Donation: A New Era of Trust and Efficiency

- **Authors:** A. R. R. Al-Bassam, F. N. Al-Dhuwaihi, M. T. Ismail (2021)
- **Description:** This paper explores the use of smart contracts combined with blockchain technology to automate and secure the organ donation process. It discusses how smart contracts can facilitate seamless transactions in organ donation, ensuring donor consent, organ matching, and recipient verification. The study suggests that this technological innovation can reduce delays, improve fairness, and increase overall trust in the organ donation system.

6.Title: Blockchain for Organ Donation and Transplantation: A Secure Approach to Increasing Organ Availability

- **Authors:** D. S. Jayaprabha, A. A. Hamed, K. R. Raju (2020)
- **Description:** This paper investigates how blockchain can be utilized to increase organ availability by improving the transparency and efficiency of organ donation systems. It suggests that blockchain can resolve issues such as organ trafficking and data manipulation, making the process more efficient and equitable. Additionally, the study discusses how blockchain can enable real-time monitoring and automated matching of organs to eligible recipients.

7.Title: Decentralized Organ Donation and Matching System Using Blockchain Technology

- **Authors:** P. N. Kumar, R. R. Manohar, K. A. Ahmed (2021)
- **Description:** This paper discusses a decentralized system for organ donation and matching using blockchain technology. It outlines how the system can securely store and share medical records, donor eligibility, and organ availability, improving the matching process and reducing delays in organ allocation. The authors propose a blockchain-based model that enhances privacy, transparency, and security in organ donation systems.

8.Title: Integrating Blockchain into Organ Donation Systems for Enhanced Transparency and Efficiency

- **Authors:** J. T. R. Scott, M. E. L. Johnson, S. R. Thompson (2021)
- **Description:** This research paper focuses on integrating blockchain into existing organ donation systems to enhance transparency, accountability, and security. The study explores the potential for using blockchain to track organ history, donor consent, and medical conditions, offering a more streamlined and reliable method for organ matching. The authors emphasize the importance of maintaining ethical standards while adopting blockchain technology in healthcare.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

The existing systems for organ donation, though well-established, face numerous challenges such as long waiting times, lack of transparency, inefficient allocation, and concerns about data privacy and security. Traditional organ donation processes are typically managed by centralized databases, government organizations, and non-governmental agencies, which store and process sensitive data related to donors, recipients, and medical conditions. These systems, however, are often criticized for being vulnerable to fraud, data manipulation, and human errors due to the reliance on centralized control. In the current state, centralized organ donation registries and matching systems are used to match available organs with recipients based on various factors like blood type, organ size, medical urgency, and geographical location.

DISADVANTAGES OF EXISITING SYSTEM

- Complexity and Implementation Challenges
- Regulatory and Legal Issues
- Privacy Concerns
- Scalability Issues
- Cost of Implementation and Maintenance

3.2 PROPOSED SYSTEM

The proposed system for organ donation using blockchain aims to create a secure, transparent, and efficient framework for managing organ donations and transplants. By leveraging blockchain's decentralized and immutable ledger, the system ensures that donor and recipient information, as well as organ matching and allocation details, are securely stored and tamper-proof. Smart contracts are employed to automate critical processes, such as verifying donor eligibility and matching organs to recipients based on medical criteria, while ensuring compliance with ethical and legal guidelines. The blockchain platform offers real-time tracking of organ transfers, enhancing transparency and accountability in the entire process. Additionally, the system provides privacy protection through encrypted data, ensuring that sensitive medical information is securely managed while still accessible to authorized parties.

ADVANTAGES OF PROPOSED SYSTEM :

- Enhanced Transparency and Trust
- Data Security and Privacy
- Improved Efficiency and Speed
- Fair and Equitable Allocation

CHAPTER 4

SYSTEM STUDY

FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are

- **ECONOMICAL FEASIBILITY**
- **TECHNICAL FEASIBILITY**
- **SOCIAL FEASIBILITY**

ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

CHAPTER 5

SYSTEM REQUIREMENTS

5.1 HARD REQUIRMENTS :

HARDWARE REQUIRMENTS :

- System : i3 or above.
- Ram : 4 GB.
- Hard Disk : 40 GB

SOFTWARE REQUIRMENTS :

- Operating system : Windows8 or Above.
- Coding Language : python

CHAPTER 6

SYSTEMDESIGN

6.1 SYSTEM ARCHITECTURE

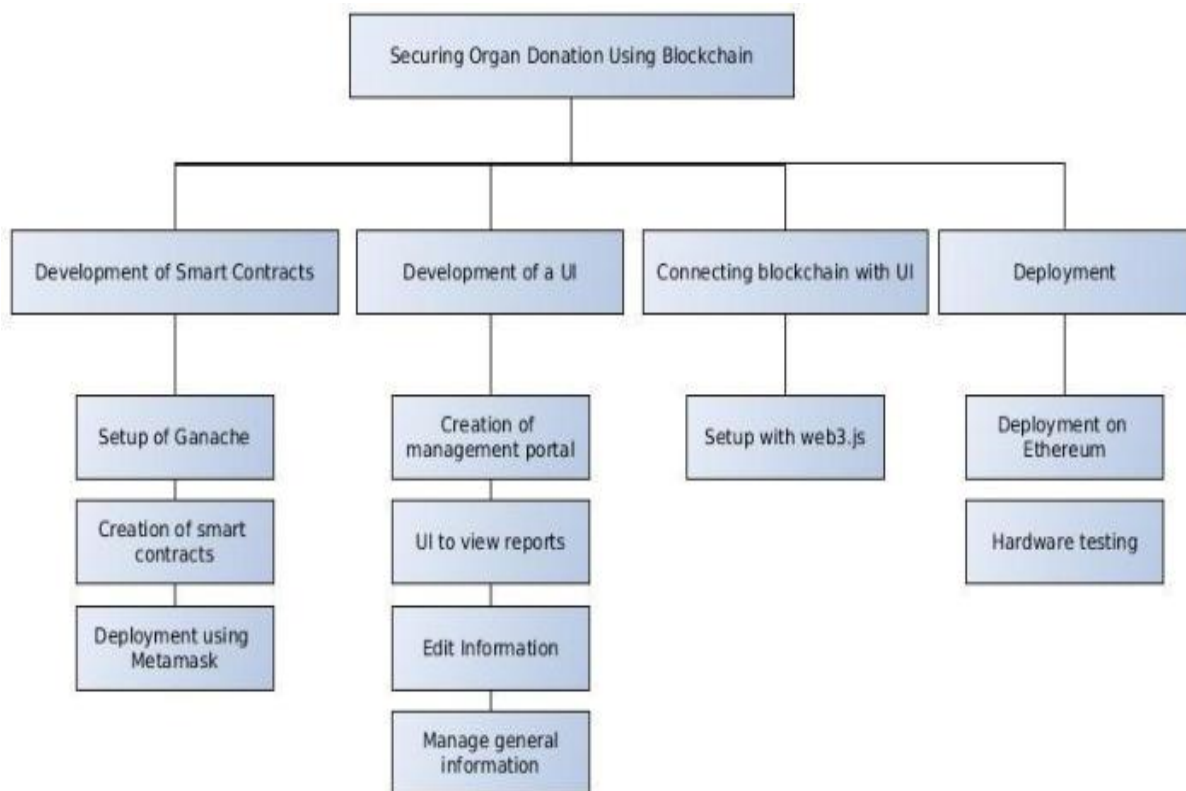


Fig 01: Architecture of System

6.2 UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group. The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML. The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems. The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems. The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

GOALS:

1. The Primary goals in the design of the UML are as follows:
2. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
3. Provide extendibility and specialization mechanisms to extend the core concepts.
4. Be independent of particular programming languages and development process.
5. Provide a formal basis for understanding the modeling language.
6. Encourage the growth of OO tools market.
7. Support higher level development concepts such as collaborations, frameworks, patterns and components.
8. Integrate best practices.

1.USECASE:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

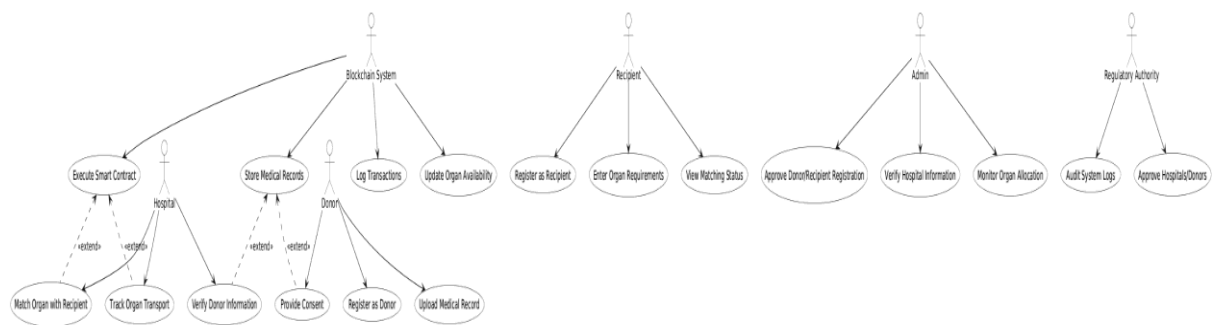


Fig 02; Organ Donation System

2. CLASS DIAGRAMS

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information

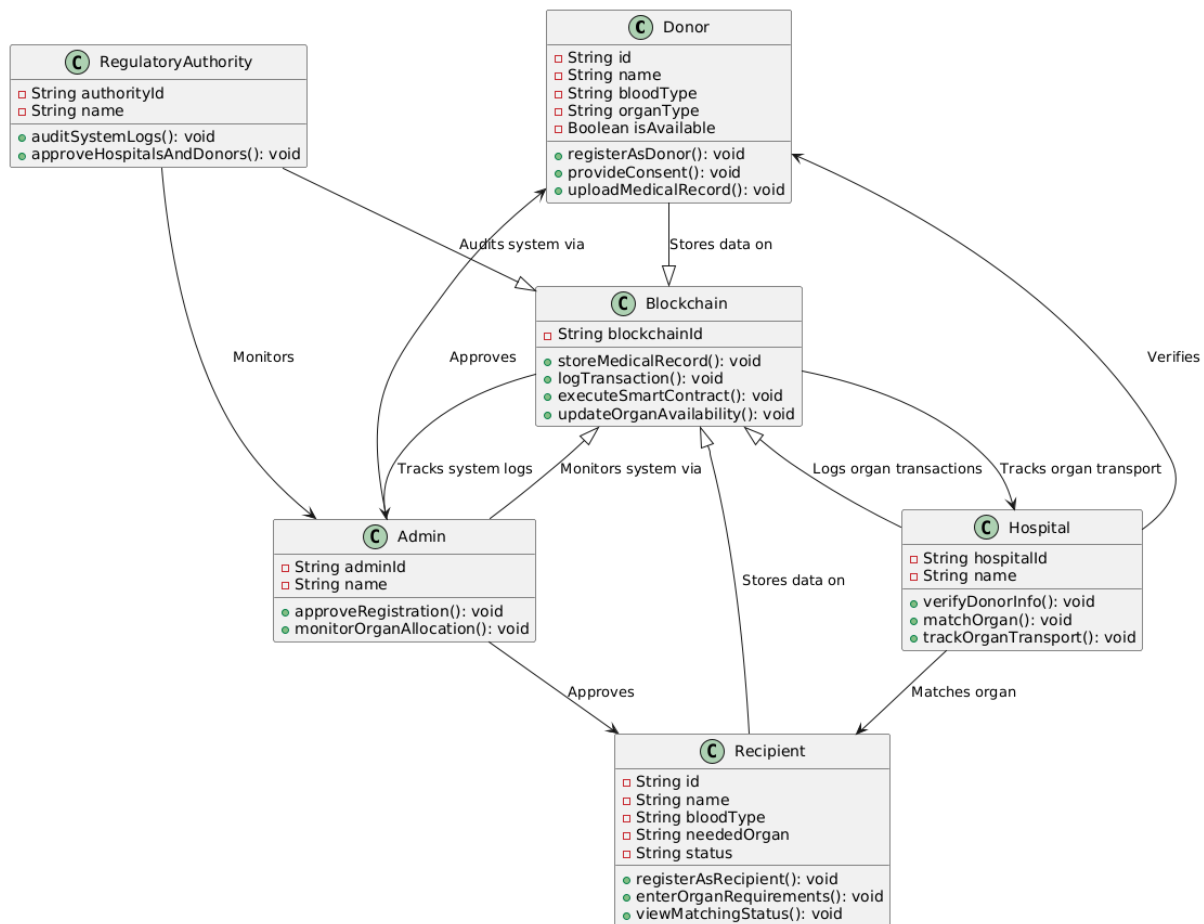


Fig 03: Class Diagram of system

3.SEQUENCE DIAGRAM

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

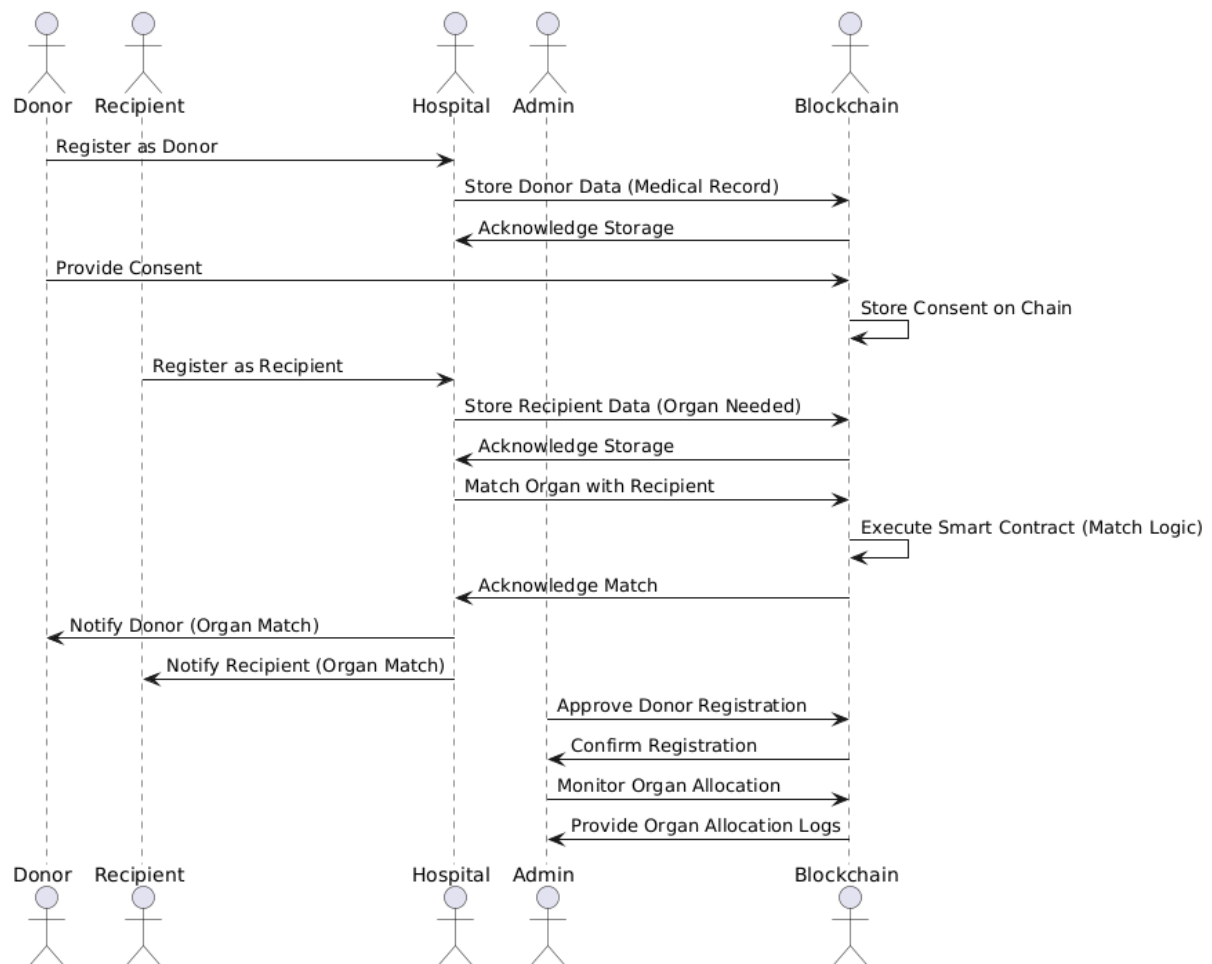


Fig 04: Sequence of UML:

4.ACTIVITY DIAGRAM:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and open rational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.

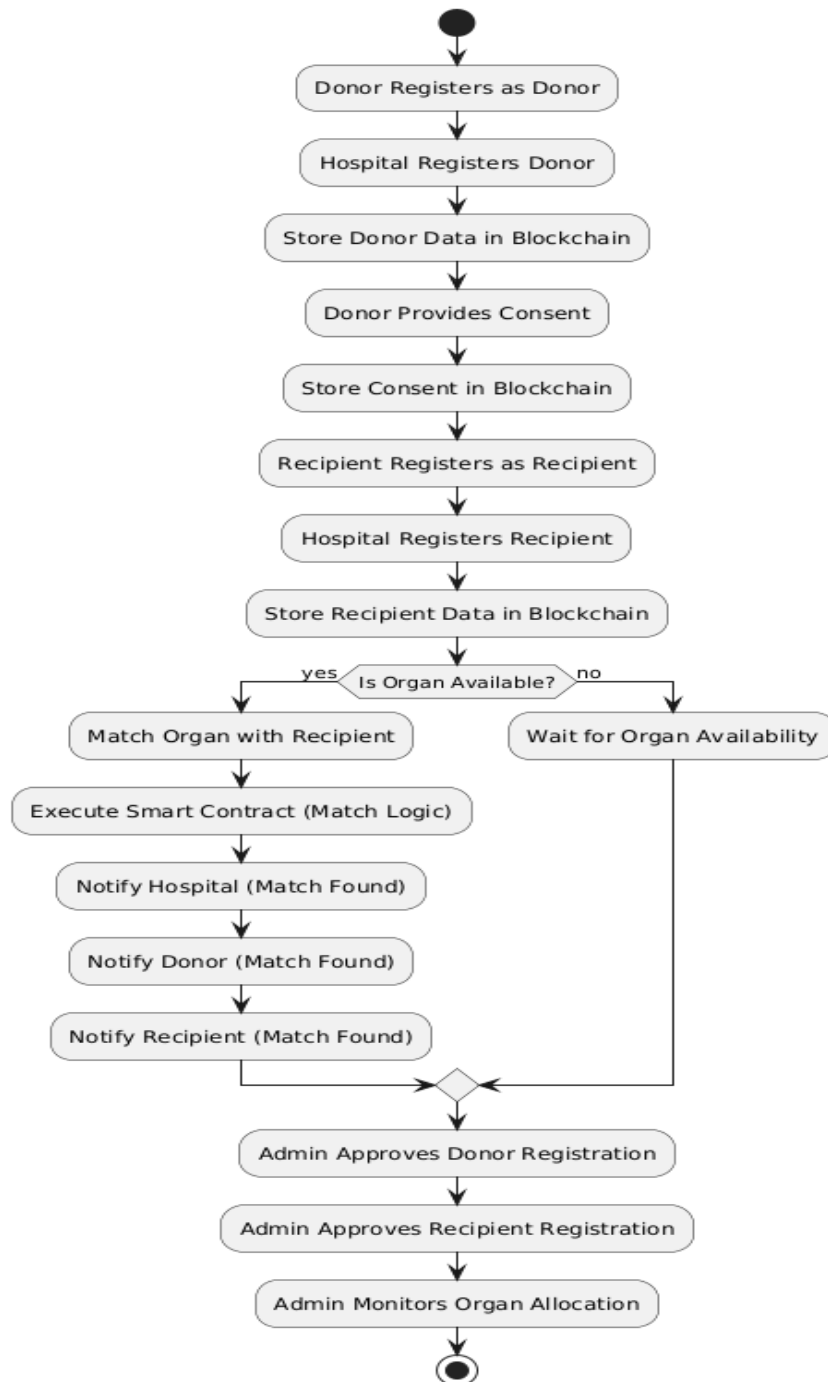


Fig 05: Flow chart of System

CHAPTER 7

IMPLEMENTATION

MODULES

1. User Registration & Authentication Module

Purpose: Enables secure onboarding of donors, recipients, hospitals, and authorities.

Features:

- Donor and recipient sign-up/login
- Hospital/authority login via verified credentials
- KYC (Know Your Customer) verification
- Biometric / 2FA support (optional)

Tech Stack:

- Frontend UI + MetaMask or Web3 Wallet
- Backend auth with JWT / OAuth
- Blockchain wallet integration

2. Donor Registration & Consent Module

Purpose: Allows users to register as organ donors and provide digital consent.

Features:

- Organ selection (kidney, heart, etc.)
- Health & blood group info submission
- Smart contract-based digital consent
- Donor availability toggle (alive, deceased)

Stored On:

- Blockchain (basic info & consent)
- IPFS (health docs)

3. Recipient Registration & Matching Module

Purpose: Allows patients to register and get matched with donors.

Features:

- Enter required organ and blood type
- Smart contract-based priority queue (urgency, compatibility, etc.)
- Automated matching system
- Notification of match

Logic Handled By:

- Smart contracts + optional AI module for enhanced matching

4. Organ Verification & Medical Record Management Module

Purpose: Secure verification and handling of medical documents.

Features:

- Upload and encrypt medical reports
- Store in IPFS with hash on blockchain
- Only verified hospitals can view/decrypt
- Audit trail for all changes

5. Organ Matching & Allocation Module

Purpose: Algorithmically match donors and recipients.

Features:

- Smart contract executes matching logic
- Matches based on organ type, blood type, urgency, and geography
- Triggers a match event on-chain

CHAPTER 8

SOFTWARE ENVIRONMENT

What is Python :

Below are some facts about Python.

Python is currently the most widely used multi-purpose, high-level programming language.

Python allows programming in Object-Oriented and Procedural paradigms. Python programs generally are smaller than other programming languages like Java.

Programmers have to type relatively less and indentation requirement of the language, makes them readable all the time.

Python language is being used by almost all tech-giant companies like – Google, Amazon, Facebook, Instagram, Dropbox, Uber... etc.

The biggest strength of Python is huge collection of standard library which can be used for the following –

- Machine Learning.
- GUI Applications (like Kivy, Tkinter, PyQt etc.)
- Web frameworks like Django (used by YouTube, Instagram, Dropbox)
- Image processing (like Opencv, Pillow)
- Web scraping (like Scrapy, BeautifulSoup, Selenium)
- Test frameworks
- Multimedia

Advantages of Python :-

Let's see how Python dominates over other languages.

1. Extensive Libraries

Python downloads with an extensive library and it contain code for various purposes like regular expressions, documentation-generation, unit-testing, web browsers, threading, databases, CGI, email, image manipulation, and more. So, we don't have to write the complete code for that manually.

2. Extensible

As we have seen earlier, Python can be extended to other languages. You can write some of your code in languages like C++ or C. This comes in handy, especially in projects.

3. Embeddable

Complimentary to extensibility, Python is embeddable as well. You can put your Python code in your source code of a different language, like C++. This lets us add scripting capabilities to our code in the other language.

4. Improved Productivity

The language's simplicity and extensive libraries render programmers **more productive** than languages like Java and C++ do. Also, the fact that you need to write less and get more things done.

5. IOT Opportunities

Since Python forms the basis of new platforms like Raspberry Pi, it finds the future bright for the Internet Of Things. This is a way to connect the language with the real world.

6. Efficiency

When working with Java, you may have to create a class to print '**Hello World**'. But in Python, just a print statement will do. It is also quite **easy** to learn, understand, and **code**. This is why when people pick up Python, they have a hard time adjusting to other more verbose languages like Java.

7. Readable

Because it is not such a verbose language, reading Python is much like reading English. This is the reason why it is so easy to learn, understand, and code. It also does not need curly braces to define blocks, and **indentation is mandatory**. This further aids the readability of the code.

8. Object-Oriented

This language supports both the procedural and object-oriented programming paradigms. While functions help us with code reusability, classes and objects let us model the real world. A class allows the **encapsulation of data** and functions into one.

9. Free and Open-Source

Like we said earlier, Python is freely available. But not only can you download python for free, but you can also download its source code, make changes to it, and even distribute it. It downloads with an extensive collection of libraries to help you with your tasks.

10. Portable

When you code your project in a language like C++, you may need to make some changes to it if you want to run it on another platform. But it isn't the same with Python. Here, you need to **code only once**, and you can run it anywhere. This is called **Write Once Run Anywhere (WORA)**. However, you need to be careful enough not to include any system-dependent features.

11. Interpreted

Lastly, we will say that it is an interpreted language. Since statements are executed one by one, **debugging is easier** than in compiled languages.

Any doubts till now in the advantages of Python? Mention in the comment section.

Advantages of Python Over Other Languages :

1. Less Coding

Almost all of the tasks done in Python requires less coding when the same task is done in other languages. Python also has an awesome standard library support, so you don't have to search for any third-party libraries to get your job done. This is the reason that many people suggest learning Python to beginners.

2. Affordable

Python is free therefore individuals, small companies or big organizations can leverage the free available resources to build applications. Python is popular and widely used so it gives you better community support.

The 2019 Github annual survey showed us that Python has overtaken Java in the most popular programming language category.

3. Python is for Everyone

Python code can run on any machine whether it is Linux, Mac or Windows. Programmers need to learn different languages for different jobs but with Python, you can professionally build web apps, perform

data analysis and machine learning, automate things, do web scraping and also build games and powerful visualizations. It is an all-rounder programming language.

Disadvantages of Python

So far, we've seen why Python is a great choice for your project. But if you choose it, you should be aware of its consequences as well. Let's now see the downsides of choosing Python over another language.

1. Speed Limitations

We have seen that Python code is executed line by line. But since python is interpreted, it often results in **slow execution**. This, however, isn't a problem unless speed is a focal point for the project. In other words, unless high speed is a requirement, the benefits offered by Python are enough to distract us from its speed limitations.

2. Weak in Mobile Computing and Browsers

While it serves as an excellent server-side language, Python is much rarely seen on the **client-side**. Besides that, it is rarely ever used to implement smartphone-based applications. One such application is called **Carbonnelle**.

The reason it is not so famous despite the existence of Brython is that it isn't that secure.

3. Design Restrictions

As you know, Python is **dynamically-typed**. This means that you don't need to declare the type of variable while writing the code. It uses **duck-typing**. But wait, what's that? Well, it just means that if it looks like a duck, it must be a duck. While this is easy on the programmers during coding, it can **raise run-time errors**.

4. Underdeveloped Database Access Layers

Compared to more widely used technologies like **JDBC (Java DataBase Connectivity)** and **ODBC (Open DataBase Connectivity)**, Python's database access layers are a bit underdeveloped. Consequently, it is less often applied in huge enterprises.

5. Simple

No, we're not kidding. Python's simplicity can indeed be a problem. Take my example. I don't do Java, I'm more of a Python person. To me, its syntax is so simple that the verbosity of Java code seems unnecessary.

This was all about the Advantages and Disadvantages of Python Programming Language.

History of Python :-

What do the alphabet and the programming language Python have in common? Right, both start with ABC. If we are talking about ABC in the Python context, it's clear that the programming language ABC is meant. ABC is a general-purpose programming language and programming environment, which had been developed in the Netherlands, Amsterdam, at the CWI (Centrum Wiskunde & Informatica). The greatest achievement of ABC was to influence the design of Python. Python was conceptualized in the late 1980s. Guido van Rossum worked that time in a project at the CWI, called Amoeba, a distributed operating system. In an interview with Bill Venners¹, Guido van Rossum said: "In the early 1980s, I worked as an implementer on a team building a language called ABC at Centrum voor Wiskunde en Informatica (CWI). I don't know how well people know ABC's influence on Python. I try to mention ABC's influence because I'm indebted to everything I learned during that project and to the people who worked on it." Later on in the same Interview, Guido van Rossum continued: "I remembered all my experience and some of my frustration with ABC. I decided to try to design a simple scripting language that possessed some of ABC's better properties, but without its problems. So I started typing. I created a simple virtual machine, a simple parser, and a simple runtime. I made my own version of the various ABC parts that I liked. I created a basic syntax, used indentation for statement grouping instead of curly braces or begin-end blocks, and developed a small number of powerful data types: a hash table (or dictionary, as we call it), a list, strings, and numbers."

What is Machine Learning :-

Before we take a look at the details of various machine learning methods, let's start by looking at what machine learning is, and what it isn't. Machine learning is often categorized as a subfield of artificial intelligence, but I find that categorization can often be misleading at first brush. The study of machine learning certainly arose from research in this context, but in the data science application of machine learning methods, it's more helpful to think of machine learning as a means of *building models of data*.

Fundamentally, machine learning involves building mathematical models to help understand data. "Learning" enters the fray when we give these models *tunable parameters* that can be adapted to observed data; in this way the program can be considered to be "learning" from the data. Once these models have been fit to previously seen data, they can be used to predict and understand aspects of newly observed data. I'll leave to the reader the more philosophical digression regarding the extent to which this type of mathematical, model-based "learning" is similar to the "learning" exhibited by the human brain. Understanding the problem setting in machine learning is essential to using these tools effectively, and so we will start with some broad categorizations of the types of approaches we'll discuss here.

Categories Of Machine Learning :-

At the most fundamental level, machine learning can be categorized into two main types: supervised learning and unsupervised learning.

Supervised learning involves somehow modeling the relationship between measured features of data and some label associated with the data; once this model is determined, it can be used to apply labels to new, unknown data. This is further subdivided into classification tasks and regression tasks: in classification, the labels are discrete categories, while in regression, the labels are continuous quantities. We will see examples of both types of supervised learning in the following section.

Unsupervised learning involves modeling the features of a dataset without reference to any label, and is often described as "letting the dataset speak for itself." These models include tasks such as clustering and dimensionality reduction. Clustering algorithms identify distinct groups of data, while dimensionality reduction algorithms search for more succinct representations of the data. We will see examples of both types of unsupervised learning in the following section.

Need for Machine Learning

Human beings, at this moment, are the most intelligent and advanced species on earth because they can think, evaluate and solve complex problems. On the other side, AI is still in its initial stage and haven't surpassed human intelligence in many aspects. Then the question is that what is the need to make machine learn? The most suitable reason for doing this is, "to make decisions, based on data, with efficiency and scale".

Lately, organizations are investing heavily in newer technologies like Artificial Intelligence, Machine Learning and Deep Learning to get the key information from data to perform several real-world tasks and solve problems. We can call it data-driven decisions taken by machines, particularly to automate the process. These data-driven decisions can be used, instead of using programming logic, in the problems that cannot be programmed inherently. The fact is that we can't do without human intelligence, but other aspect is that we all need to solve real-world problems with efficiency at a huge scale. That is why the need for machine learning arises.

Challenges in Machines Learning :-

While Machine Learning is rapidly evolving, making significant strides with cybersecurity and autonomous cars, this segment of AI as whole still has a long way to go. The reason behind is that ML has not been able to overcome number of challenges. The challenges that ML is facing currently are –

Quality of data – Having good-quality data for ML algorithms is one of the biggest challenges. Use of low-quality data leads to the problems related to data preprocessing and feature extraction.

Time-Consuming task – Another challenge faced by ML models is the consumption of time especially for data acquisition, feature extraction and retrieval.

Lack of specialist persons – As ML technology is still in its infancy stage, availability of expert resources is a tough job.

No clear objective for formulating business problems – Having no clear objective and well-defined goal for business problems is another key challenge for ML because this technology is not that mature yet.

Issue of overfitting & underfitting – If the model is overfitting or underfitting, it cannot be represented well for the problem.

Curse of dimensionality – Another challenge ML model faces is too many features of data points. This can be a real hindrance.

Difficulty in deployment – Complexity of the ML model makes it quite difficult to be deployed in real life.

Applications of Machines Learning :-

Machine Learning is the most rapidly growing technology and according to researchers we are in the golden year of AI and ML. It is used to solve many real-world complex problems which cannot be solved with traditional approach. Following are some real-world applications of ML –

- Emotion analysis
 - Sentiment analysis
 - Error detection and prevention
 - Weather forecasting and prediction
 - Stock market analysis and forecasting
 - Speech synthesis
 - Speech recognition
 - Customer segmentation
 - Object recognition
 - Fraud detection
 - Fraud prevention
 - Recommendation of products to customer in online shopping
- How to Start Learning Machine Learning?

Arthur Samuel coined the term “**Machine Learning**” in 1959 and defined it as a “**Field of study that gives computers the capability to learn without being explicitly programmed**”.

And that was the beginning of Machine Learning! In modern times, Machine Learning is one of the most popular (if not the most!) career choices. According to indeed, Machine Learning Engineer Is The Best Job of 2019 with a 344% growth and an average base salary of **\$146,085** per year.

But there is still a lot of doubt about what exactly is Machine Learning and how to start learning it? So this article deals with the Basics of Machine Learning and also the path you can follow to eventually become a full-fledged Machine Learning Engineer. Now let's get started!!!

How to start learning ML?

This is a rough roadmap you can follow on your way to becoming an insanely talented Machine Learning Engineer. Of course, you can always modify the steps according to your needs to reach your desired end-goal!

Step 1 – Understand the Prerequisites

In case you are a genius, you could start ML directly but normally, there are some prerequisites that you need to know which include Linear Algebra, Multivariate Calculus, Statistics, and Python. And if you don't know these, never fear! You don't need a Ph.D. degree in these topics to get started but you do need a basic understanding.

(a) Learn Linear Algebra and Multivariate Calculus

Both Linear Algebra and Multivariate Calculus are important in Machine Learning. However, the extent to which you need them depends on your role as a data scientist. If you are more focused on application heavy machine learning, then you will not be that heavily focused on maths as there are many common libraries available. But if you want to focus on R&D in Machine Learning, then mastery of Linear Algebra and Multivariate Calculus is very important as you will have to implement many ML algorithms from scratch.

(b) Learn Statistics

Data plays a huge role in Machine Learning. In fact, around 80% of your time as an ML expert will be spent collecting and cleaning data. And statistics is a field that handles the collection, analysis, and presentation of data. So it is no surprise that you need to learn it!!!

Some of the key concepts in statistics that are important are Statistical Significance, Probability Distributions, Hypothesis Testing, Regression, etc. Also, Bayesian Thinking is also a very important part of ML which deals with various concepts like Conditional Probability, Priors, and Posteriors, Maximum Likelihood, etc.

(c) Learn Python

Some people prefer to skip Linear Algebra, Multivariate Calculus and Statistics and learn them as they go along with trial and error. But the one thing that you absolutely cannot skip is python! While there are other languages you can use for Machine Learning like R, Scala, etc. Python is currently the most popular language for ML. In fact, there are many Python libraries that are specifically useful for Artificial Intelligence and Machine Learning such as keras, tensorflow, scikit learn, etc.

So if you want to learn ML, it's best if you learn Python! You can do that using various online resources and courses such as fork python available Free on GeeksforGeeks.

Step 2 – Learn Various ML Concepts

Now that you are done with the prerequisites, you can move on to actually learning ML (Which is the fun part!!!) It's best to start with the basics and then move on to the more complicated stuff. Some of the basic concepts in ML are:

(a) Terminologies of Machine Learning

- **Model** – A model is a specific representation learned from data by applying some machine learning algorithm. A model is also called a hypothesis.
- **Feature** – A feature is an individual measurable property of the data. A set of numeric features can be conveniently described by a feature vector. Feature vectors are fed as input to the model. For example, in order to predict a fruit, there may be features like color, smell, taste, etc.
- **Target (Label)** – A target variable or label is the value to be predicted by our model. For the fruit example discussed in the feature section, the label with each set of input would be the name of the fruit like apple, orange, banana, etc.
- **Training** – The idea is to give a set of inputs(features) and it's expected outputs(labels), so after training, we will have a model (hypothesis) that will then map new data to one of the categories trained on.
- **Prediction** – Once our model is ready, it can be fed a set of inputs to which it will provide a predicted output(label).

(b) Types of Machine Learning

- **Supervised Learning** – This involves learning from a training dataset with labeled data using classification and regression models. This learning process continues until the required level of performance is achieved.

- **Unsupervised Learning** – This involves using unlabelled data and then finding the underlying structure in the data in order to learn more and more about the data itself using factor and cluster analysis models.
- **Semi-supervised Learning** – This involves using unlabelled data like Unsupervised Learning with a small amount of labeled data. Using labeled data vastly increases the learning accuracy and is also more cost-effective than Supervised Learning.
- **Reinforcement Learning** – This involves learning optimal actions through trial and error. So the next action is decided by learning behaviors that are based on the current state and that will maximize the reward in the future.

Advantages of Machine learning :-

1. Easily identifies trends and patterns -

Machine Learning can review large volumes of data and discover specific trends and patterns that would not be apparent to humans. For instance, for an e-commerce website like Amazon, it serves to understand the browsing behaviors and purchase histories of its users to help cater to the right products, deals, and reminders relevant to them. It uses the results to reveal relevant advertisements to them.

2. No human intervention needed (automation)

With ML, you don't need to babysit your project every step of the way. Since it means giving machines the ability to learn, it lets them make predictions and also improve the algorithms on their own. A common example of this is anti-virus softwares ; they learn to filter new threats as they are recognized. ML is also good at recognizing spam.

3. Continuous Improvement

As ML algorithms gain experience, they keep improving in accuracy and efficiency. This lets them make better decisions. Say you need to make a weather forecast model. As the amount of data you have keeps growing, your algorithms learn to make more accurate predictions faster.

4. Handling multi-dimensional and multi-variety data

Machine Learning algorithms are good at handling data that are multi-dimensional and multi-variety, and they can do this in dynamic or uncertain environments.

5. Wide Applications

You could be an e-tailer or a healthcare provider and make ML work for you. Where it does apply, it holds the capability to help deliver a much more personal experience to customers while also targeting the right customers.

Disadvantages of Machine Learning :-

1. Data Acquisition

Machine Learning requires massive data sets to train on, and these should be inclusive/unbiased, and of good quality. There can also be times where they must wait for new data to be generated.

2. Time and Resources

ML needs enough time to let the algorithms learn and develop enough to fulfill their purpose with a considerable amount of accuracy and relevancy. It also needs massive resources to function. This can mean additional requirements of computer power for you.

3. Interpretation of Results

Another major challenge is the ability to accurately interpret results generated by the algorithms. You must also carefully choose the algorithms for your purpose.

4. High error-susceptibility

Machine Learning is autonomous but highly susceptible to errors. Suppose you train an algorithm with data sets small enough to not be inclusive. You end up with biased predictions coming from a biased training set. This leads to irrelevant advertisements being displayed to customers. In the case of ML, such blunders can set off a chain of errors that can go undetected for long periods of time. And when they do get noticed, it takes quite some time to recognize the source of the issue, and even longer to correct it.

Python Development Steps :-

Guido Van Rossum published the first version of Python code (version 0.9.0) at alt.sources in February 1991. This release included already exception handling, functions, and the core data types of list, dict, str and others. It was also object oriented and had a module system. Python version 1.0 was released in January 1994. The major new features included in this release were the functional programming tools

lambda, map, filter and reduce, which Guido Van Rossum never liked. Six and a half years later in October 2000, Python 2.0 was introduced. This release included list comprehensions, a full garbage collector and it was supporting unicode. Python flourished for another 8 years in the versions 2.x before the next major release as Python 3.0 (also known as "Python 3000" and "Py3K") was released. Python 3 is not backwards compatible with Python 2.x. The emphasis in Python 3 had been on the removal of duplicate programming constructs and modules, thus fulfilling or coming close to fulfilling the 13th law of the Zen of Python: "There should be one -- and preferably only one -- obvious way to do it." Some changes in Python 3.0:

- Print is now a function
- Views and iterators instead of lists
- The rules for ordering comparisons have been simplified. E.g. a heterogeneous list cannot be sorted, because all the elements of a list must be comparable to each other.
- There is only one integer type left, i.e. int. long is int as well.
- The division of two integers returns a float instead of an integer. "/" can be used to have the "old" behaviour.
- Text Vs. Data Instead Of Unicode Vs. 8-bit

Purpose :-

We demonstrated that our approach enables successful segmentation of intra-retinal layers—even with low-quality images containing speckle noise, low contrast, and different intensity ranges throughout—with the assistance of the ANIS feature.

Python

Python is an interpreted high-level programming language for general-purpose programming. Created by Guido van Rossum and first released in 1991, Python has a design philosophy that emphasizes code readability, notably using significant whitespace.

Python features a dynamic type system and automatic memory management. It supports multiple programming paradigms, including object-oriented, imperative, functional and procedural, and has a large and comprehensive standard library.

- Python is Interpreted – Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- Python is Interactive – you can actually sit at a Python prompt and interact with the interpreter directly to write your programs.

Python also acknowledges that speed of development is important. Readable and terse code is part of this, and so is access to powerful constructs that avoid tedious repetition of code. Maintainability also ties into this may be an all but useless metric, but it does say something about how much code you have to scan, read and/or understand to troubleshoot problems or tweak behaviors. This speed of development, the ease with which a programmer of other languages can pick up basic Python skills and the huge standard library is key to another area where Python excels. All its tools have been quick to implement, saved a lot of time, and several of them have later been patched and updated by people with no Python background - without breaking.

Modules Used in Project :-

Tensorflow

TensorFlow is a free and open-source software library for dataflow and differentiable programming across a range of tasks. It is a symbolic math library, and is also used for machine learning applications such as neural networks. It is used for both research and production at Google.

TensorFlow was developed by the Google Brain team for internal Google use. It was released under the Apache 2.0 open-source license on November 9, 2015.

Numpy

Numpy is a general-purpose array-processing package. It provides a high-performance multidimensional array object, and tools for working with these arrays.

It is the fundamental package for scientific computing with Python. It contains various features including these important ones:

- A powerful N-dimensional array object
- Sophisticated (broadcasting) functions
- Tools for integrating C/C++ and Fortran code

- Useful linear algebra, Fourier transform, and random number capabilities

Besides its obvious scientific uses, Numpy can also be used as an efficient multi-dimensional container of generic data. Arbitrary data-types can be defined using Numpy which allows Numpy to seamlessly and speedily integrate with a wide variety of databases.

Pandas

Pandas is an open-source Python Library providing high-performance data manipulation and analysis tool using its powerful data structures. Python was majorly used for data munging and preparation. It had very little contribution towards data analysis. Pandas solved this problem. Using Pandas, we can accomplish five typical steps in the processing and analysis of data, regardless of the origin of data load, prepare, manipulate, model, and analyze. Python with Pandas is used in a wide range of fields including academic and commercial domains including finance, economics, Statistics, analytics, etc.

Matplotlib

Matplotlib is a Python 2D plotting library which produces publication quality figures in a variety of hardcopy formats and interactive environments across platforms. Matplotlib can be used in Python scripts, the Python and IPython shells, the Jupyter Notebook, web application servers, and four graphical user interface toolkits. Matplotlib tries to make easy things easy and hard things possible. You can generate plots, histograms, power spectra, bar charts, error charts, scatter plots, etc., with just a few lines of code. For examples, see the sample plots and [thumbnail_gallery](#).

For simple plotting the pyplot module provides a MATLAB-like interface, particularly when combined with IPython. For the power user, you have full control of line styles, font properties, axes properties, etc, via an object oriented interface or via a set of functions familiar to MATLAB users.

Scikit – learn

Scikit-learn provides a range of supervised and unsupervised learning algorithms via a consistent interface in Python. It is licensed under a permissive simplified BSD license and is distributed under many Linux distributions, encouraging academic and commercial use. **Python**

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Install Python Step-by-Step in Windows and Mac :

Python a versatile programming language doesn't come pre-installed on your computer devices. Python was first released in the year 1991 and until today it is a very popular high-level programming language. Its style philosophy emphasizes code readability with its notable use of great whitespace.

The object-oriented approach and language construct provided by Python enables programmers to write both clear and logical code for projects. This software does not come pre-packaged with Windows.

How to Install Python on Windows and Mac :

There have been several updates in the Python version over the years. The question is how to install Python? It might be confusing for the beginner who is willing to start learning Python but this tutorial will solve your query. The latest or the newest version of Python is version 3.7.4 or in other words, it is Python

Note: The python version 3.7.4 cannot be used on Windows XP or earlier devices.

Before you start with the installation process of Python. First, you need to know about your **System Requirements**. Based on your system type i.e. operating system and based processor, you must download the python version. My system type is a **Windows 64-bit operating system**. So the steps below are to install python version 3.7.4 on Windows 7 device or to install Python 3. Download the Python Cheat sheet here. The steps on how to install Python on Windows 10, 8 and 7 are **divided into 4 parts** to help understand better.

Download the Correct version into the system

Step 1: Go to the official site to download and install python using Google Chrome or any other web browser. OR Click on the following link: <https://www.python.org>



Fig 06: Accessing the site for installing the py.

Now, check for the latest and the correct version for your operating system.

Step 2: Click on the Download Tab.



Fig 07: By clicking the ‘Download’

Step 3: You can either select the Download Python for windows 3.7.4 button in Yellow Color or you can scroll further down and click on download with respective to their version. Here, we are downloading the most recent python version for windows 3.7.4

Looking for a specific release?

Python releases by version number:

Release version	Release date		Click for more
Python 3.7.4	July 8, 2019	 Download	Release Notes
Python 3.6.9	July 2, 2019	 Download	Release Notes
Python 3.7.3	March 25, 2019	 Download	Release Notes
Python 3.4.10	March 18, 2019	 Download	Release Notes
Python 3.5.7	March 18, 2019	 Download	Release Notes
Python 2.7.16	March 4, 2019	 Download	Release Notes
Python 3.7.2	Dec. 24, 2018	 Download	Release Notes

Fig 08: Version of py

Step 4: Scroll down the page until you find the Files option.

Step 5: Here you see a different version of python along with the operating system.

Files					
Version	Operating System	Description	MD5 Sum	File Size	GPC
Gzipped source tarball	Source release		68111671e5b2db4aef7b9ab010f09be	23017663	50G
XZ compressed source tarball	Source release		d33e4a8e6097051c2eca45ee3604803	17131432	50G
macOS 64-bit/32-bit installer	Mac OS X	for Mac OS X 10.8 and later	6428b4fa7583daf1a440c8a8ce08e6	34896416	50G
macOS 64-bit installer	Mac OS X	for OS X 10.9 and later	5dd605c38217a45773b75e4a936b243f	20082845	50G
Windows help file	Windows		d63999573a2c56b2ac5fca6e6b47cd2	8131761	50G
Windows x86-64 embeddable zip file	Windows	for AMD64/EM64/x64	9800c3cf9d9ec0b9ab6f3184a40729a2	7304391	50G
Windows x86-64 executable installer	Windows	for AMD64/EM64/x64	a702b4bca076d9bdc3043a383e563400	2688368	50G
Windows x86-64 web-based installer	Windows	for AMD64/EM64/x64	28cb1c6088d72a8be53a3b351b4bd2	1362904	50G
Windows x86 embeddable zip file	Windows		9fab3b8178b41879fda94123574129d8	6741626	50G
Windows x86 executable installer	Windows		23cc802942a5446a3d645147e394789	25663848	50G
Windows x86 web-based installer	Windows		1b670cfa5d317df82c30983ea371897c	1324608	50G

Fig 09 ; OS supports Syytem

- To download Windows 32-bit python, you can select any one from the three options: Windows x86 embeddable zip file, Windows x86 executable installer or Windows x86 web-based installer.
- To download Windows 64-bit python, you can select any one from the three options: Windows x86-64 embeddable zip file, Windows x86-64 executable installer or Windows x86-64 web-based installer.

Here we will install Windows x86-64 web-based installer. Here your first part regarding which version of python is to be downloaded is completed. Now we move ahead with the second part in installing python i.e. Installation

Note: To know the changes or updates that are made in the version you can click on the Release Note Option.

Installation of Python

Step 1: Go to Download and Open the downloaded python version to carry out the installation process.

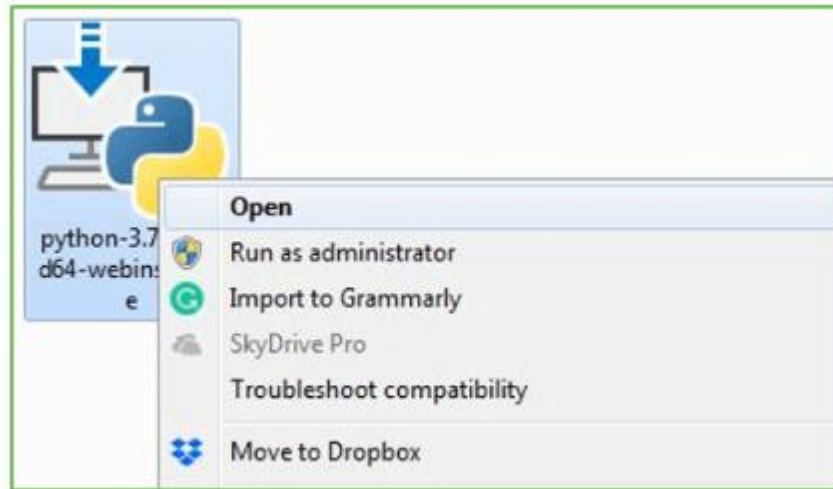


Fig 10: Run as Administrator.

Step 2: Before you click on Install Now, Make sure to put a tick on Add Python 3.7 to PATH.



Fig 11: Set up of py

Step 3: Click on Install NOW After the installation is successful. Click on Close.



Fig 12: Full set up

With these above three steps on python installation, you have successfully and correctly installed Python. Now is the time to verify the installation.

Note: The installation process might take a couple of minutes.

Verify the Python Installation

Step 1: Click on Start

Step 2: In the Windows Run Command, type “cmd”.

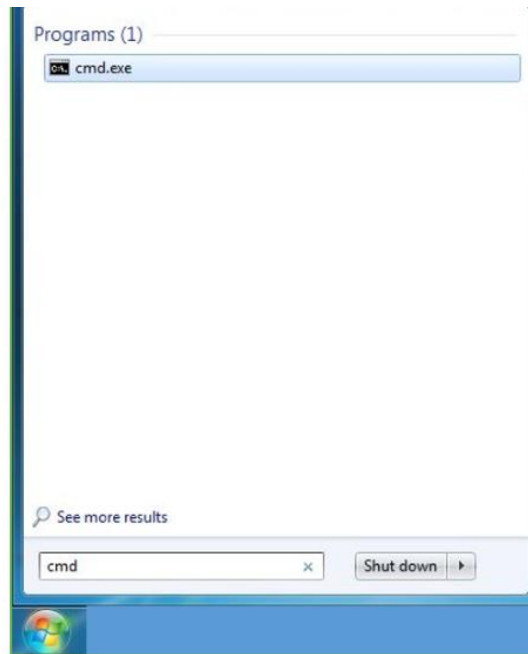


Fig 13: CMD

Step 3: Open the Command prompt option.

Step 4: Let us test whether the python is correctly installed. Type **python -V** and press Enter.

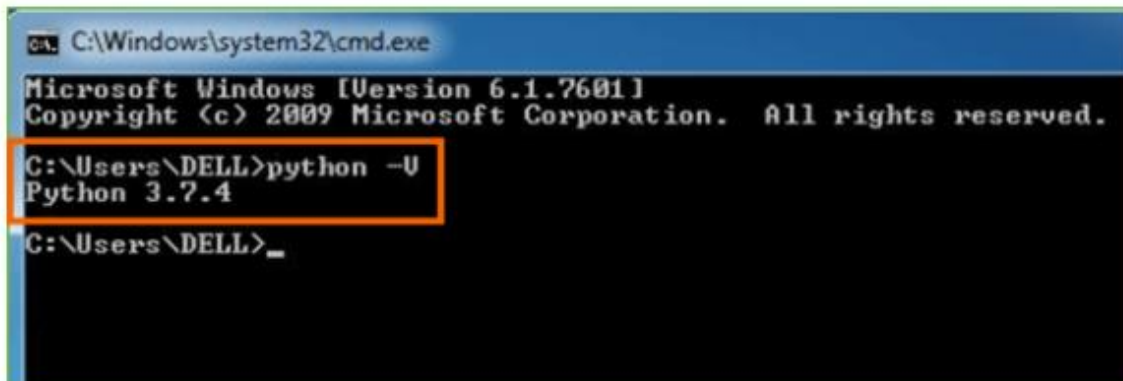


Fig 14 : checking the version

Step 5: You will get the answer as 3.7.4

Note: If you have any of the earlier versions of Python already installed. You must first uninstall the earlier version and then install the new one.

Check how the Python IDLE works

Step 1: Click on Start

Step 2: In the Windows Run command, type “python idle”.

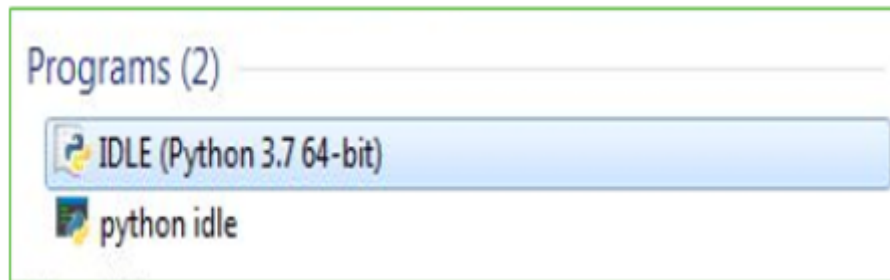


Fig 15: IDLE python with 64-bit

Step 3: Click on IDLE (Python 3.7 64-bit) and launch the program

Step 4: To go ahead with working in IDLE you must first save the file. **Click on File > Click on Save**

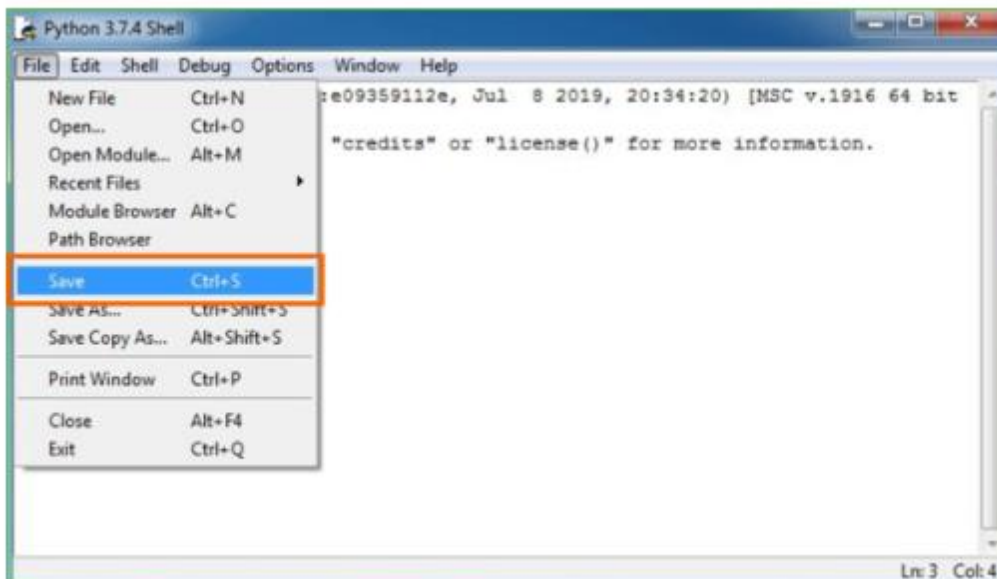


Fig 16: Shortcut to sav

Step 5: Name the file and save as type should be Python files. Click on SAVE. Here I have named the files as Hey World.

Step 6: Now for e.g. **enter print**

CHAPTER 9

SOURCE CODE

SAMPLE CODE

MANAGE.PY

```
#!/usr/bin/env python

import os

import sys


if __name__ == '__main__':

    os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'Organ.settings')

    try:

        from django.core.management import execute_from_command_line

    except ImportError as exc:

        raise ImportError(

            "Couldn't import Django. Are you sure it's installed and "

            "available on your PYTHONPATH environment variable? Did you "

            "forget to activate a virtual environment?"

        ) from exc

    execute_from_command_line(sys.argv)
```

SETTINGS.PY

```
"""
```

Django settings for Organ project.

Generated by 'django-admin startproject' using Django 2.1.7.

For more information on this file, see

<https://docs.djangoproject.com/en/2.1/topics/settings/>

For the full list of settings and their values, see

<https://docs.djangoproject.com/en/2.1/ref/settings/>

"""

import os

Build paths inside the project like this: os.path.join(BASE_DIR, ...)

BASE_DIR = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))

Quick-start development settings - unsuitable for production

See <https://docs.djangoproject.com/en/2.1/howto/deployment/checklist/>

SECURITY WARNING: keep the secret key used in production secret!

SECRET_KEY = 'x*84rnhc55fvut#6hlj)-u*^01c6_!!itml+ahgl+q4=b7s0i'

SECURITY WARNING: don't run with debug turned on in production!

DEBUG = True

```
ALLOWED_HOSTS = []
```

```
# Application definition
```

```
INSTALLED_APPS = [
```

```
    'django.contrib.admin',
```

```
    'django.contrib.auth',
```

```
    'django.contrib.contenttypes',
```

```
    'django.contrib.sessions',
```

```
    'django.contrib.messages',
```

```
    'django.contrib.staticfiles',
```

```
    'OrganApp'
```

```
]
```

```
MIDDLEWARE = [
```

```
    'django.middleware.security.SecurityMiddleware',
```

```
    'django.contrib.sessions.middleware.SessionMiddleware',
```

```
    'django.middleware.common.CommonMiddleware',
```

```
    'django.middleware.csrf.CsrfViewMiddleware',
```

```
    'django.contrib.auth.middleware.AuthenticationMiddleware',
```

```
    'django.contrib.messages.middleware.MessageMiddleware',
```

```
'django.middleware.clickjacking.XFrameOptionsMiddleware',  
  
]  
  
ROOT_URLCONF = 'Organ.urls'  
  
TEMPLATES = [  
  
    {  
  
        'BACKEND': 'django.template.backends.django.DjangoTemplates',  
  
        'DIRS': [],  
  
        'APP_DIRS': True,  
  
        'OPTIONS': {  
  
            'context_processors': [  
  
                'django.template.context_processors.debug',  
  
                'django.template.context_processors.request',  
  
                'django.contrib.auth.context_processors.auth',  
  
                'django.contrib.messages.context_processors.messages',  
  
            ],  
  
        },  
  
    ],  
  
]  
  
WSGI_APPLICATION = 'Organ.wsgi.application'
```

```
# Database
```

```
# https://docs.djangoproject.com/en/2.1/ref/settings/#databases
```

```
DATABASES = {
```

```
    'default': {
```

```
        'ENGINE': 'django.db.backends.sqlite3',
```

```
        'NAME': os.path.join(BASE_DIR, 'db.sqlite3'),
```

```
    }
```

```
}
```

```
# Password validation
```

```
# https://docs.djangoproject.com/en/2.1/ref/settings/#auth-password-validators
```

```
AUTH_PASSWORD_VALIDATORS = [
```

```
    {
```

```
        'NAME': 'django.contrib.auth.password_validation.UserAttributeSimilarityValidator',
```

```
    },
```

```
    {
```

```
        'NAME': 'django.contrib.auth.password_validation.MinimumLengthValidator',
```

```
    },
```

```
    {
```

```
        'NAME': 'django.contrib.auth.password_validation.CommonPasswordValidator',

    },

    {

        'NAME': 'django.contrib.auth.password_validation.NumericPasswordValidator',

    },

]
```

Internationalization

<https://docs.djangoproject.com/en/2.1/topics/i18n/>

LANGUAGE_CODE = 'en-us'

TIME_ZONE = 'UTC'

USE_I18N = True

USE_L10N = True

USE_TZ = True

Static files (CSS, JavaScript, Images)

<https://docs.djangoproject.com/en/2.1/howto/static-files/>

```
STATIC_URL = '/static/'
```

SOURCE CODE:

VIEWS.PY

```
from django.shortcuts import render

from django.template import RequestContext

from django.contrib import messages

import json

from web3 import Web3, HTTPProvider

from django.http import HttpResponse

from django.core.files.storage import FileSystemStorage

import os

import random

from datetime import date


global user, hospital


def readDetails(contract_type):

    global details

    details = ""

    print(contract_type+"=====")

    blockchain_address = 'http://127.0.0.1:9545' #Blokchain connection IP
```



```

web3 = Web3(HTTPProvider(blockchain_address))

web3.eth.defaultAccount = web3.eth.accounts[0]

compiled_contract_path = 'Organ.json' #Organ contract code

deployed_contract_address = '0xd374Cb05bd6187D6cF905D7bBD85f2b704fBDD29' #hash address
to access Organ contract

with open(compiled_contract_path) as file:

    contract_json = json.load(file) # load contract info as JSON

    contract_abi = contract_json['abi'] # fetch contract's abi - necessary to call its functions

file.close()

contract = web3.eth.contract(address=deployed_contract_address, abi=contract_abi) #now calling
contract to access data

if contract_type == 'signup':

    details = contract.functions.getUser().call()

if contract_type == 'donor':

    details = contract.functions.getDonor().call()

if contract_type == 'patient':

    details = contract.functions.getPatient().call()

print(details)

def saveDataBlockChain(currentData, contract_type):

    global details

    global contract

    details = ""

```

```

blockchain_address = 'http://127.0.0.1:9545'

web3 = Web3(HTTPProvider(blockchain_address))

web3.eth.defaultAccount = web3.eth.accounts[0]

compiled_contract_path = 'Organ.json' #Organ contract file

    deployed_contract_address = '0xd374Cb05bd6187D6cF905D7bBD85f2b704fBDD29' #contract
address

with open(compiled_contract_path) as file:

    contract_json = json.load(file) # load contract info as JSON

    contract_abi = contract_json['abi'] # fetch contract's abi - necessary to call its functions

file.close()

contract = web3.eth.contract(address=deployed_contract_address, abi=contract_abi)

readDetails(contract_type)

if contract_type == 'signup':

    details+=currentData

    msg = contract.functions.addUser(details).transact()

    tx_receipt = web3.eth.waitForTransactionReceipt(msg)

if contract_type == 'donor':

    details+=currentData

    msg = contract.functions.setDonor(details).transact()

    tx_receipt = web3.eth.waitForTransactionReceipt(msg)

if contract_type == 'patient':

    details+=currentData

    msg = contract.functions.setPatient(details).transact()

```

```

tx_receipt = web3.eth.waitForTransactionReceipt(msg)

def updateConsent(currentData, contract_type):

    blockchain_address = 'http://127.0.0.1:9545'

    web3 = Web3(HTTPProvider(blockchain_address))

    web3.eth.defaultAccount = web3.eth.accounts[0]

    compiled_contract_path = 'Organ.json' #organ contract file

    deployed_contract_address = '0xd374Cb05bd6187D6cF905D7bBD85f2b704fBDD29' #contract
address

    with open(compiled_contract_path) as file:

        contract_json = json.load(file) # load contract info as JSON

        contract_abi = contract_json['abi'] # fetch contract's abi - necessary to call its functions

    file.close()

    contract = web3.eth.contract(address=deployed_contract_address, abi=contract_abi)

    if contract_type == "donor":

        msg = contract.functions.setDonor(currentData).transact()

        tx_receipt = web3.eth.waitForTransactionReceipt(msg)

    else:

        msg = contract.functions.setPatient(currentData).transact()

        tx_receipt = web3.eth.waitForTransactionReceipt(msg)

def Alert(request):

    if request.method == 'GET':

```

```

global user

pid = request.GET.get('pid', False)

did = request.GET.get('did', False)

record = ""

readDetails("patient")

rows = details.split("\n")

for i in range(len(rows)-1):

    arr = rows[i].split("#")

    if arr[0] != pid:

        record += rows[i]+"\n"

    else:

record += arr[0]+"#"+arr[1]+"#"+arr[2]+"#"+arr[3]+"#"+arr[4]+"#"+arr[5]+"#"+arr[6]+"#"+arr[7]+"#"+arr[8]+"#
Donor "+did+" Matched\n"

updateConsent(record, "patient")

record = ""

readDetails("donor")

rows = details.split("\n")

for i in range(len(rows)-1):

    arr = rows[i].split("#")

    if arr[0] != did:

        record += rows[i]+"\n"

```

```

else:

record +=
arr[0]+"#"+arr[1]+"#"+arr[2]+"#"+arr[3]+"#"+arr[4]+"#"+arr[5]+"#"+arr[6]+"#"+arr[7]+"#"+arr[8]+"#
Patients "+pid+" Matched\n"

updateConsent(record, "donor")

context= {'data':"Alert Sent to both Patinet : "+pid+" & Donor : "+did+" About Matched"}

return render(request, 'HospitalScreen.html', context)


def MatchOrganAction(request):

    if request.method == 'GET':

        global user

        pid = request.GET.get('pid', False)

        organs = request.GET.get('organs', False)

        columns = ['Donor ID', 'Donor Name', 'Address', 'Contact No', 'Health Condition', 'Donating Organs',
'Aadhar No', 'Hospital', 'Entry Date', 'Alert User & Donor About Matching']

        output = "<table border=1 align=center>"

        font = '<font size="" color="black">'

        for i in range(len(columns)):

            output += '<th>'+font+columns[i]+'</th>'

        output += "</tr>"

        readDetails("donor")

        rows = details.split("\n")

```

```

for i in range(len(rows)-1):

    arr = rows[i].split("#")

    if arr[9] == 'Pending':

        output+='<tr><td>'+font+str(arr[0])+'</td>'

        output+='<td>'+font+str(arr[1])+'</td>'

        output+='<td>'+font+str(arr[2])+'</td>'

        output+='<td>'+font+str(arr[3])+'</td>'

        output+='<td>'+font+str(arr[4])+'</td>'

        output+='<td>'+font+str(arr[5])+'</td>'

        output+='<td>'+font+str(arr[6])+'</td>'

        output+='<td>'+font+str(arr[7])+'</td>'

        output+='<td>'+font+str(arr[8])+'</td>'

        output+='<td><a href=\'Alert?pid='+str(pid)+'&did='+str(arr[0])+'\'><font size=3
color=black>Send Alert</font></a></td></tr>'

    output += "<tr></tr><tr></tr><tr></tr><tr></tr><tr></tr><tr></tr>"

    context= {'data':output}

    return render(request, 'HospitalScreen.html', context)

```

```

def MatchOrgans(request):

    if request.method == 'GET':

        columns = ['Patient ID', 'Patient Name', 'Address', 'Contact No', 'Disease History', 'Required Organs',
'Aadhar No', 'Hospital', 'Entry Date', 'Match Organs']

```

```

output = "<table border=1 align=center>"

font = '<font size="" color="black">'

for i in range(len(columns)):

    output += '<th>'+font+columns[i]+'</th>'

output += "</tr>"

readDetails("patient")

rows = details.split("\n")

for i in range(len(rows)-1):

    arr = rows[i].split("#")

    if arr[9] == 'Pending':

        output+='<tr><td>'+font+str(arr[0])+'</td>'

        output+='<td>'+font+str(arr[1])+'</td>'

        output+='<td>'+font+str(arr[2])+'</td>'

        output+='<td>'+font+str(arr[3])+'</td>'

        output+='<td>'+font+str(arr[4])+'</td>'

        output+='<td>'+font+str(arr[5])+'</td>'

        output+='<td>'+font+str(arr[6])+'</td>'

        output+='<td>'+font+str(arr[7])+'</td>'

        output+='<td>'+font+str(arr[8])+'</td>'

        output+='<td><a href=\'MatchOrganAction?pid='+str(arr[0])+'&organs='+str(arr[5])+'\'><font
size=3 color=black>Click Here to Match Organs</font></a></td></tr>'

    output += "<tr></tr><tr></tr><tr></tr><tr></tr><tr></tr><tr></tr>"

context= {'data':output}

```

```

return render(request, 'HospitalScreen.html', context)

def ViewTransplant(request):

    if request.method == 'GET':

        columns = ['Patient ID', 'Patient Name', 'Address', 'Contact No', 'Disease History', 'Required Organs',
'Aadhar No', 'Hospital', 'Entry Date', 'Transplant Status']

        output = "<table border=1 align=center>"

        font = '<font size="" color="black">'

        for i in range(len(columns)):

            output += '<th>'+font+columns[i]+'</th>'

        output += "</tr>"

        readDetails("patient")

        rows = details.split("\n")

        for i in range(len(rows)-1):

            arr = rows[i].split("#")

            if arr[9] != 'Pending':

                output+='<tr><td>'+font+str(arr[0])+'</td>'

                output+='<td>'+font+str(arr[1])+'</td>'

                output+='<td>'+font+str(arr[2])+'</td>'

                output+='<td>'+font+str(arr[3])+'</td>'

                output+='<td>'+font+str(arr[4])+'</td>'

                output+='<td>'+font+str(arr[5])+'</td>'

                output+='<td>'+font+str(arr[6])+'</td>'

```



```

        output+='<td>'+font+str(arr[7])+'</td>'

        output+='<td>'+font+str(arr[8])+'</td>'

        output+='<td>'+font+str(arr[9])+'</td></tr>'

output += "<tr></tr><tr></tr><tr></tr><tr></tr><tr></tr>"

context= {'data':output}

return render(request, 'HospitalScreen.html', context)

```

```
def ViewRequestStatus(request):
```

```
    if request.method == 'GET':
```

```
        global user
```

```
        columns = ['Patient ID', 'Patient Name', 'Address', 'Contact No', 'Disease History', 'Required Organs',
'Aadhar No', 'Hospital', 'Entry Date', 'Transplant Status']
```

```
        output = "<table border=1 align=center>"
```

```
        font = '<font size="" color="black">'
```

```
        for i in range(len(columns)):
```

```
            output += '<th>'+font+columns[i]+'</th>'
```

```
        output += "</tr>"
```

```
        readDetails("patient")
```

```
        rows = details.split("\n")
```

```
        for i in range(len(rows)-1):
```

```
            arr = rows[i].split("#")
```

```
            if user == arr[0]:
```

```
                output+='<tr><td>'+font+str(arr[0])+'</td>'

```

```

        output+='<td>'+font+str(arr[1])+</td>'

        output+='<td>'+font+str(arr[2])+</td>'

        output+='<td>'+font+str(arr[3])+</td>'

        output+='<td>'+font+str(arr[4])+</td>'

        output+='<td>'+font+str(arr[5])+</td>'

        output+='<td>'+font+str(arr[6])+</td>'

        output+='<td>'+font+str(arr[7])+</td>'

        output+='<td>'+font+str(arr[8])+</td>'

        output+='<td>'+font+str(arr[9])+</td></tr>'

output += "<tr></tr><tr></tr><tr></tr><tr></tr><tr></tr>"

context= {'data':output}

return render(request, 'UserScreen.html', context)

```

```
def DonationStatus(request):
```

```
    if request.method == 'GET':
```

```
        global user
```

```
        pid = request.GET.get('pid', False)
```

```
        organs = request.GET.get('organs', False)
```

```
        columns = ['Donor ID', 'Donor Name', 'Address', 'Contact No', 'Health Condition', 'Donating Organs',
'Aadhar No', 'Hospital', 'Entry Date', 'Match Status']
```

```
        output = "<table border=1 align=center>"
```

```
        font = '<font size="" color="black">'
```

```
        for i in range(len(columns)):
```

```

        output += '<th>'+font+columns[i]+'</th>'

output += "</tr>"

readDetails("donor")

rows = details.split("\n")

for i in range(len(rows)-1):

    arr = rows[i].split("#")

    if user == arr[0]:

        output+='<tr><td>'+font+str(arr[0])+'</td>'

        output+='<td>'+font+str(arr[1])+'</td>'

        output+='<td>'+font+str(arr[2])+'</td>'

        output+='<td>'+font+str(arr[3])+'</td>'

        output+='<td>'+font+str(arr[4])+'</td>'

        output+='<td>'+font+str(arr[5])+'</td>'

        output+='<td>'+font+str(arr[6])+'</td>'

        output+='<td>'+font+str(arr[7])+'</td>'

        output+='<td>'+font+str(arr[8])+'</td>'

        output+='<td>'+font+str(arr[9])+'</td></tr>'

output += "<tr></tr><tr></tr><tr></tr><tr></tr><tr></tr>"

context= {'data':output}

return render(request, 'DonorScreen.html', context)

```

```

def AddDonorHistoryAction(request):

```

```

if request.method == 'POST':

    global hospital

    donor = request.POST.get('t1', False)

    address = request.POST.get('t2', False)

    contact = request.POST.get('t3', False)

    condition = request.POST.get('t4', False)

    organs = request.POST.get('t5', False)

    aadhar = request.POST.get('t6', False)

    today = date.today()

    readDetails("donor")

    rows = details.split("\n")

    did = len(rows)

    if did == 0:

        did = 1

    donor_id = donor+"-"+str(did)

    data = donor_id+"#"+donor+"#"+address+"#"+contact+"#"+condition+"#"+organs+"#"+aadhar+"#"+hospital+"#"+str(today)+"#Pending\n"

    saveDataBlockchain(data,"donor")

    context= {'data':'Donor Details Added with ID : '+donor_id+"<br/>Use above Donor Id for login"}

    return render(request, 'AddDonorHistory.html', context)

def AddDonorHistory(request):

```

```

if request.method == 'GET':

    return render(request, 'AddDonorHistory.html', {})

def AddPatientHistoryAction(request):

    if request.method == 'POST':

        global hospital

        patient = request.POST.get('t1', False)

        address = request.POST.get('t2', False)

        contact = request.POST.get('t3', False)

        disease = request.POST.get('t4', False)

        organs = request.POST.get('t5', False)

        aadhar = request.POST.get('t6', False)

        today = date.today()

        readDetails("patient")

        rows = details.split("\n")

        pid = len(rows)

        if pid == 0:

            pid = 1

        patient_id = patient+"-"+str(pid)

        data = patient_id+"#"+patient+"#"+address+"#"+contact+"#"+disease+"#"+organs+"#"+aadhar+"#"+hospital+"#"+str(today)+"#Pending\n"

        saveDataBlockChain(data,"patient")

```

```
context= {'data':'Patient Details Added with ID : '+patient_id+"<br/>Use above Patient Id for login"}
```

```
return render(request, 'AddPatientHistory.html', context)
```

```
def AddPatientHistory(request):
```

```
    if request.method == 'GET':
```

```
        return render(request, 'AddPatientHistory.html', {})
```

```
def index(request):
```

```
    if request.method == 'GET':
```

```
        return render(request, 'index.html', {})
```

```
def HospitalLogin(request):
```

```
    if request.method == 'GET':
```

```
        return render(request, 'HospitalLogin.html', {})
```

```
def Register(request):
```

```
    if request.method == 'GET':
```

```
        return render(request, 'Register.html', {})
```

```
def DonorLogin(request):
```

```
    if request.method == 'GET':
```

```
        return render(request, 'DonorLogin.html', {})
```

```
def UserLogin(request):

    if request.method == 'GET':

        return render(request, 'UserLogin.html', {})


def checkUser(username):

    flag = False

    readDetails("signup")

    rows = details.split("\n")

    for i in range(len(rows)-1):

        arr = rows[i].split("#")

        if arr[0] == username:

            flag = True

            break

    return flag


def Signup(request):

    if request.method == 'POST':

        username = request.POST.get('username', False)

        password = request.POST.get('password', False)

        contact = request.POST.get('contact', False)

        email = request.POST.get('email', False)
```

```

address = request.POST.get('address', False)

hospital = request.POST.get('hospital', False)

if checkUser(username) == False:

    data = username+"#" +password+"#" +contact+"#" +email+"#" +address+"#" +hospital+"\n"

    saveDataBlockchain(data,"signup")

    context= {'data':'Signup Process Completed'}

    return render(request, 'Register.html', context)

else:

    context= {'data':'Given username already exists'}

    return render(request, 'Register.html', context)

def HospitalLoginAction(request):

    if request.method == 'POST':

        global user, hospital

        username = request.POST.get('username', False)

        password = request.POST.get('password', False)

        status = 'none'

        readDetails("signup")

        rows = details.split("\n")

        for i in range(len(rows)-1):

            arr = rows[i].split("#")

            if arr[0] == username and arr[1] == password:

```



```

        status = 'success'

        hospital = arr[5]

        user = username

    if status == 'success':

        output = 'Welcome '+username

        context= {'data':output}

        return render(request, "HospitalScreen.html", context)

    if status == 'none':

        context= {'data':'Invalid login details'}

        return render(request, 'HospitalLogin.html', context)


def DonorLoginAction(request):

    if request.method == 'POST':

        global user

        username = request.POST.get('username', False)

        readDetails("donor")

        rows = details.split("\n")

        status = "none"

        for i in range(len(rows)-1):

            arr = rows[i].split("#")

            if arr[0] == username:

```

```

        status = 'success'

        user = username

    if status == 'success':

        output = 'Welcome '+username

        context= {'data':output}

        return render(request, "DonorScreen.html", context)

    if status == 'none':

        context= {'data':'Invalid login details'}

        return render(request, 'DonorLogin.html', context)

def UserLoginAction(request):

    if request.method == 'POST':

        global user

        username = request.POST.get('username', False)

        readDetails("patient")

        rows = details.split("\n")

        status = "none"

        for i in range(len(rows)-1):

            arr = rows[i].split("#")

            if arr[0] == username:

                status = 'success'

                user = username

```

```
if status == 'success':  
  
    output = 'Welcome '+username  
  
    context= {'data':output}  
  
    return render(request, "UserScreen.html", context)  
  
if status == 'none':  
  
    context= {'data':'Invalid login details'}  
  
    return render(request, 'UserLogin.html', context)
```

CHAPTER 10

SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

11.1 TYPES OF TESTS

Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfactory, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures : interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

White Box Testing

White Box Testing is a testing in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is used to test areas that cannot be reached from a black box level.

Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.



Fig:17: flow of execution

When applied to machine learning models, black box testing would mean testing machine learning models without knowing the internal details such as features of the machine learning model, the algorithm used to create the model etc. The challenge, however, is to verify the test outcome against the expected values that are known beforehand.

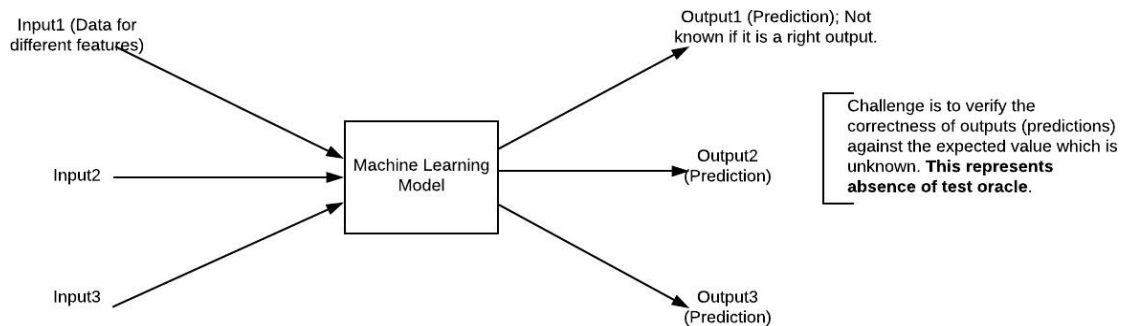


Fig 18: Model of ML

Unit Testing

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed

- All links should take the user to the correct page.

Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered

CHAPTER 11

OUTPUT SCREENS

To run project double click on 'runServer.bat' file to start python web server and get below page

```
C:\Windows\system32\cmd.exe
E:\takeoff\Dec23\OrganMatching>python manage.py runserver
Performing system checks...

System check identified no issues (0 silenced).

You have 15 unapplied migration(s). Your project may not work properly until you apply the migrations for app(s): admin, auth, contenttypes, sessions.
Run 'python manage.py migrate' to apply them.
December 26, 2023 - 21:19:24
Django version 2.1.7, using settings 'Organ.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.
```

In above screen python server started and now open browser and enter URL as <http://127.0.0.1:8000/index.html> and press enter key to get below page

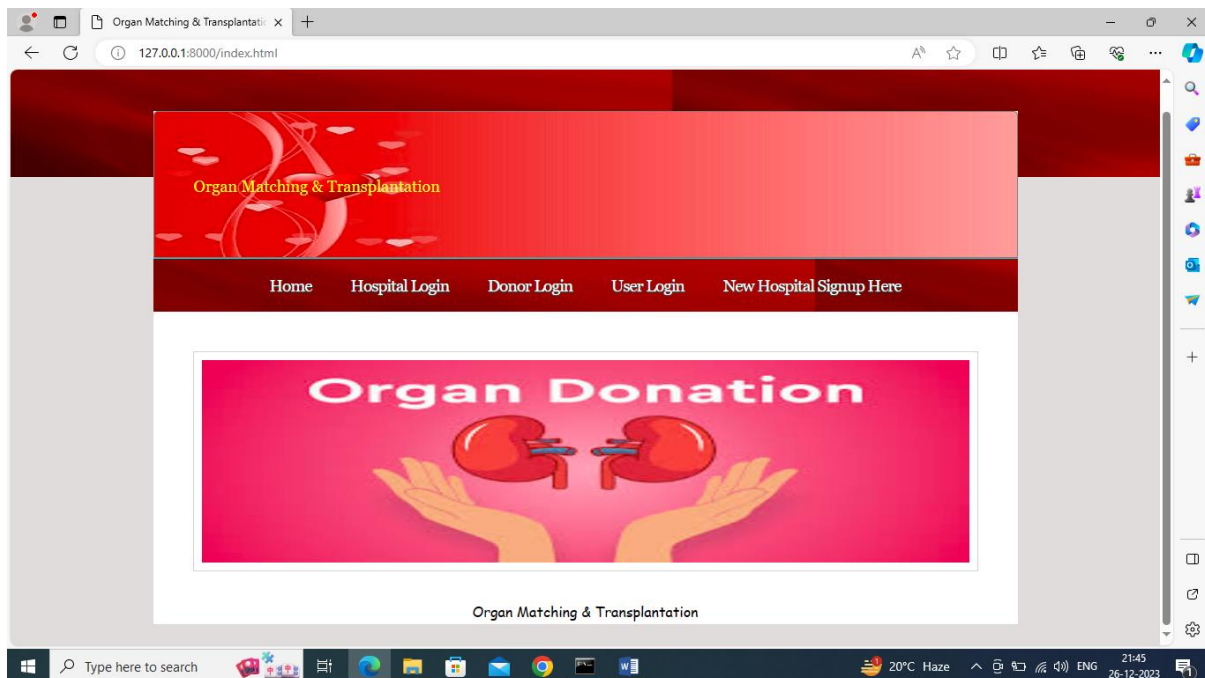


Fig 19: In above screen click on 'New Hospital Signup Here' link to get below page

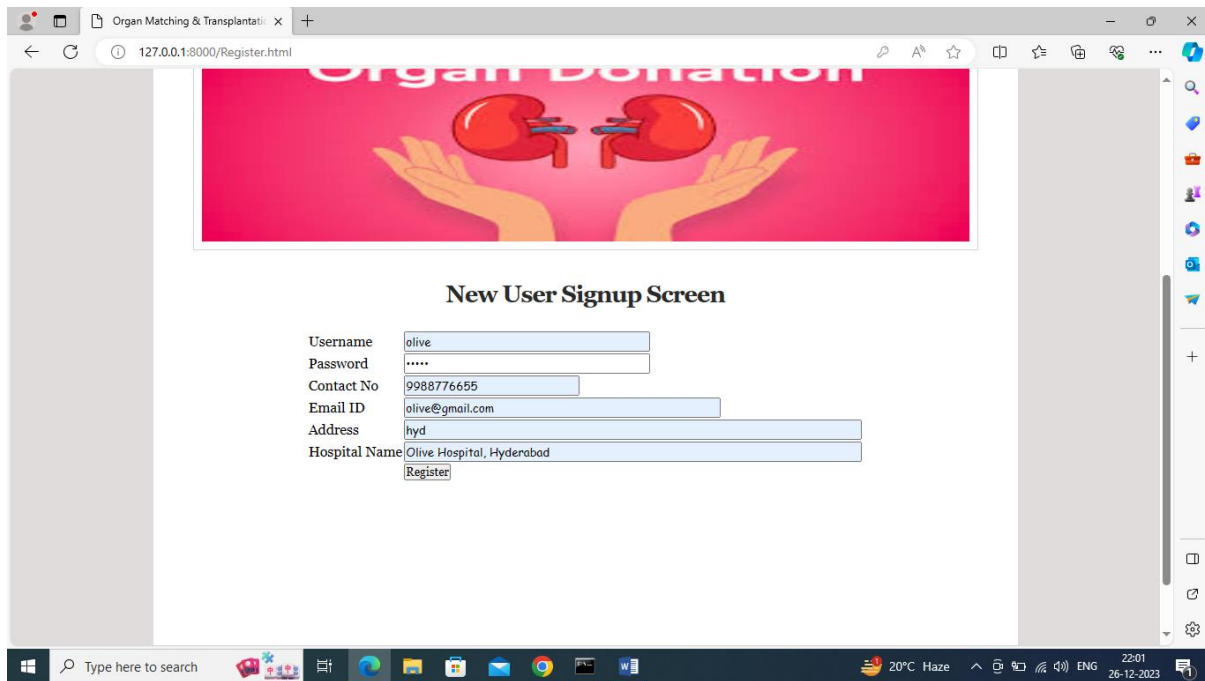


Fig 20:In above screen hospital is entering signup details and then press button to get below page

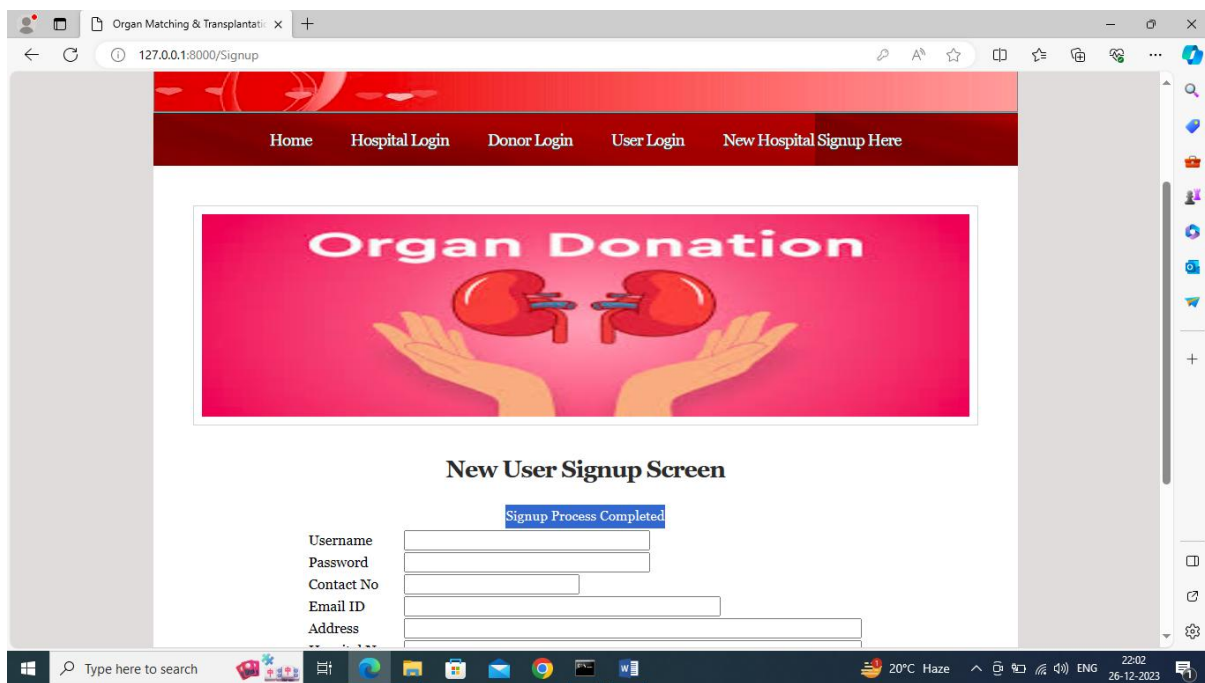


Fig 21:In above screen hospital sign up task completed and now click on 'Hospital Login' link to get below page

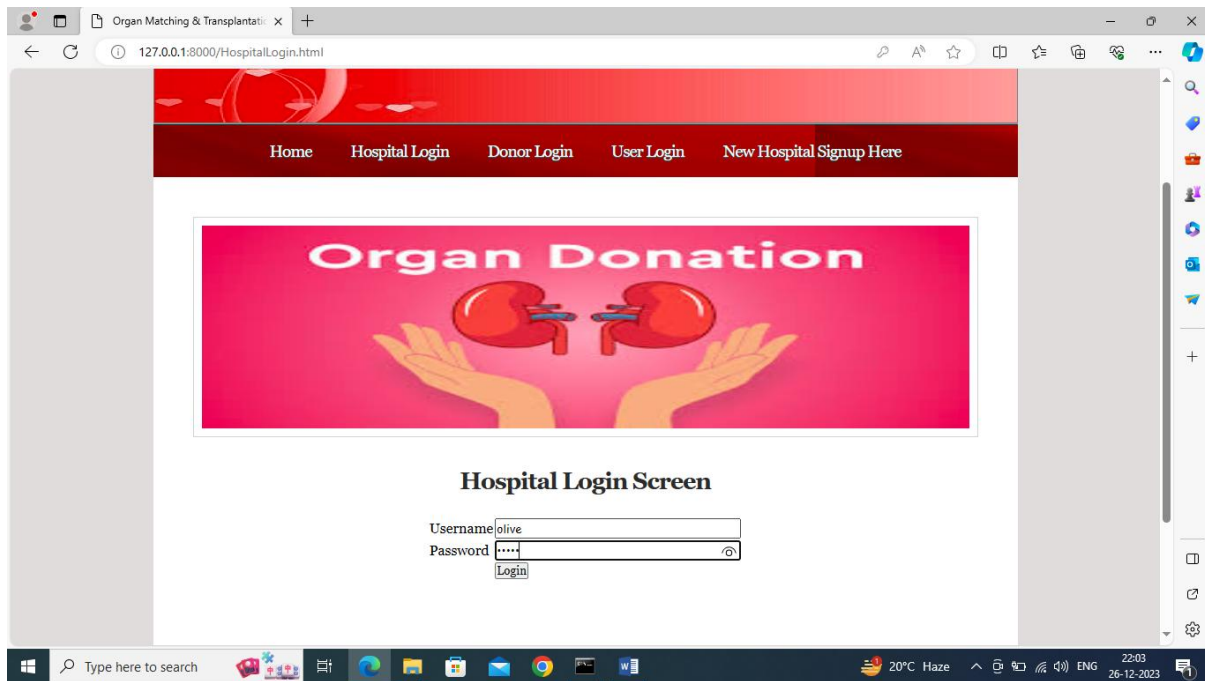


Fig 22: In above screen hospital is login and after login will get below page

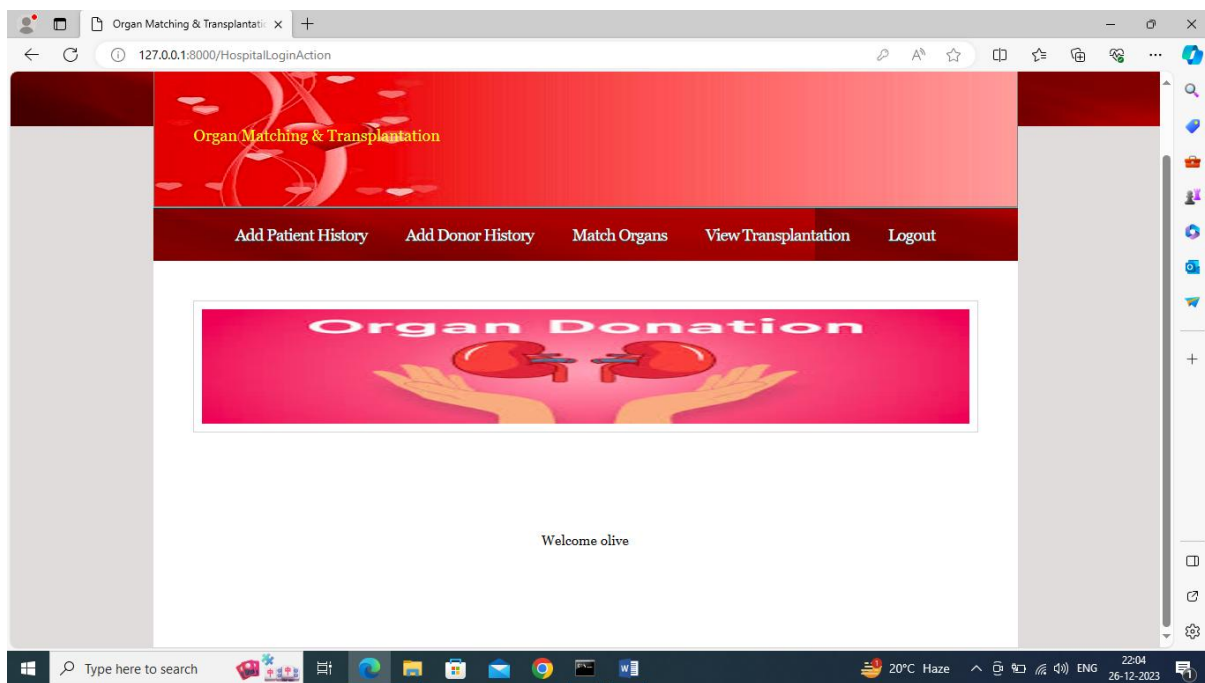
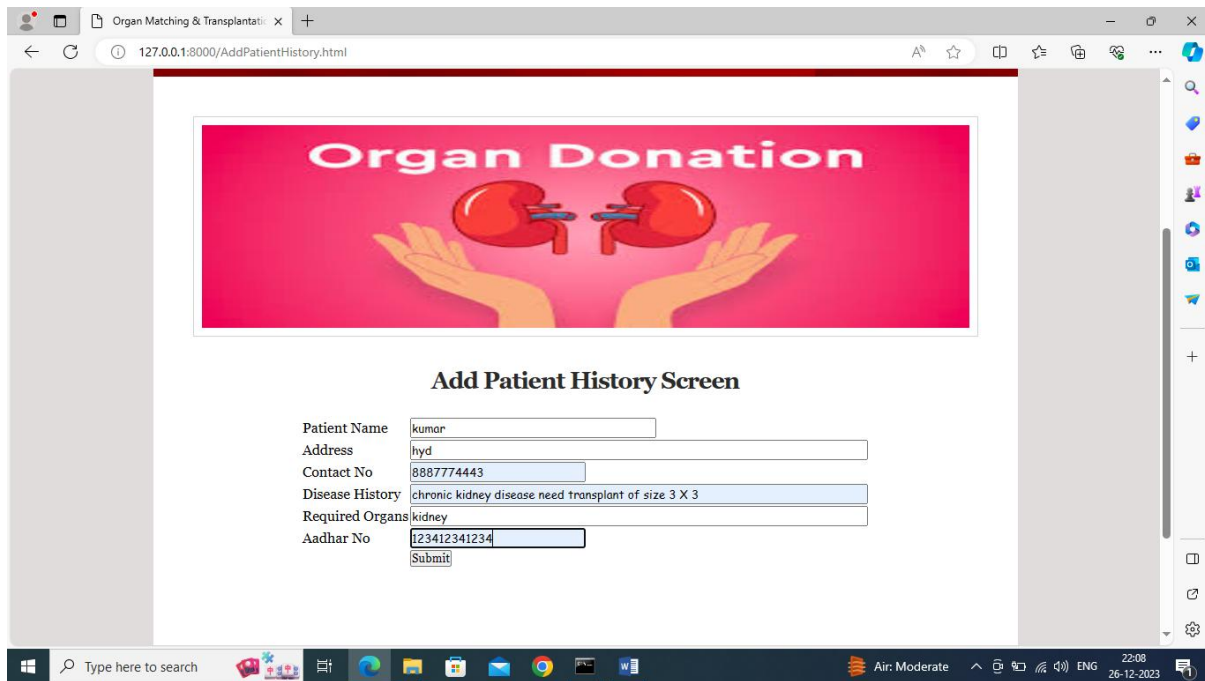


Fig 23: In above screen hospital can click on 'Add Patient History' link to add patient history for organs request

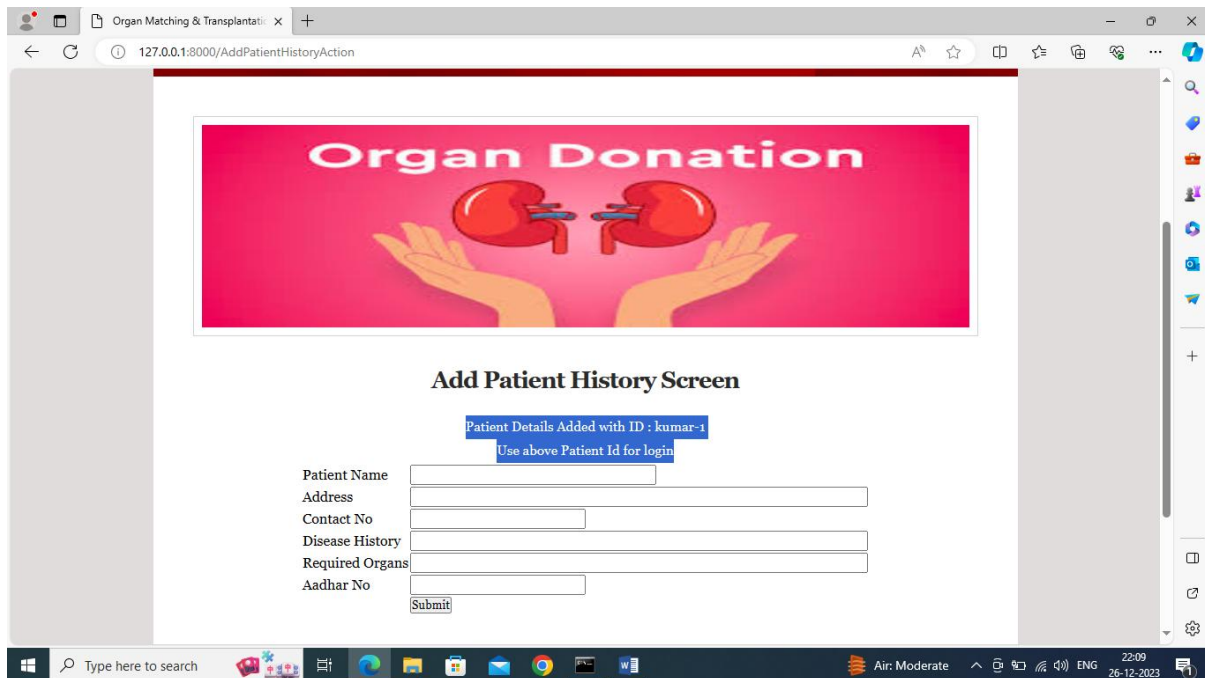


Organ Donation

Add Patient History Screen

Patient Name: kumar
 Address: hyd
 Contact No: 8887774443
 Disease History: chronic kidney disease need transplant of size 3 X 3
 Required Organs: kidney
 Aadhar No: 123412341234
 Submit

Fig 24: In above screen hospital can add patient details who need organs and all this details will be saved inside Blockchain and will get below page



Organ Donation

Add Patient History Screen

Patient Details Added with ID : kumar-1
 Use above Patient Id for login

Patient Name:
 Address:
 Contact No:
 Disease History:
 Required Organs:
 Aadhar No:
 Submit

Fig 25: In above screen patient details added with ID and this id can be used by patients to track organ matching status and similarly you can add N numbers of patients and now click on 'Add Donor History' link to get below page

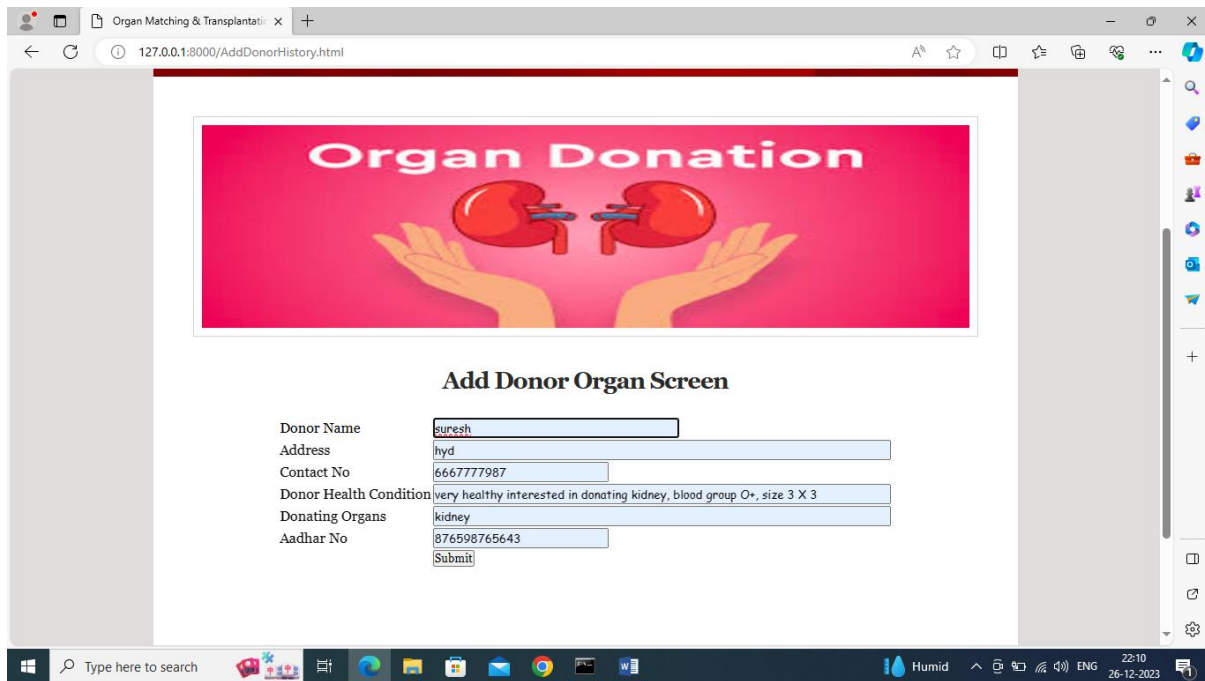


Fig 26: In above screen hospital adding donor details and now click on 'Submit' button to get below page

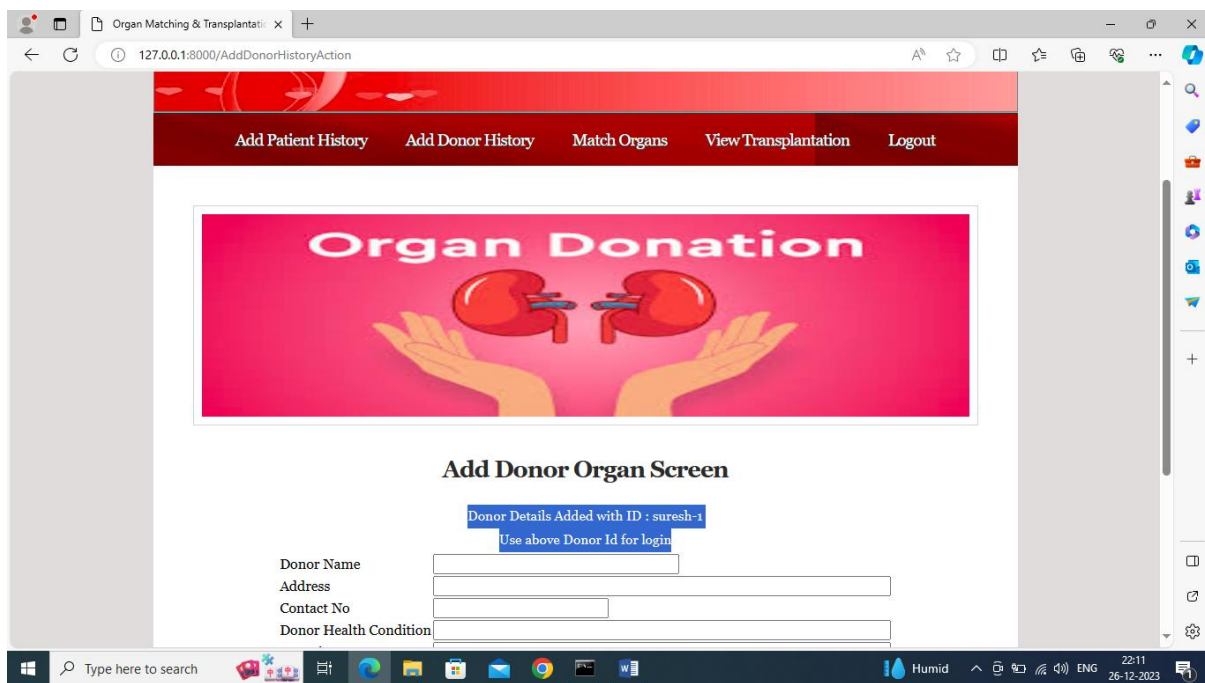


Fig 27: In above screen donating details added to Blockchain and donor can use above ID to track matching status and now click on 'Match Organs' link to get below page

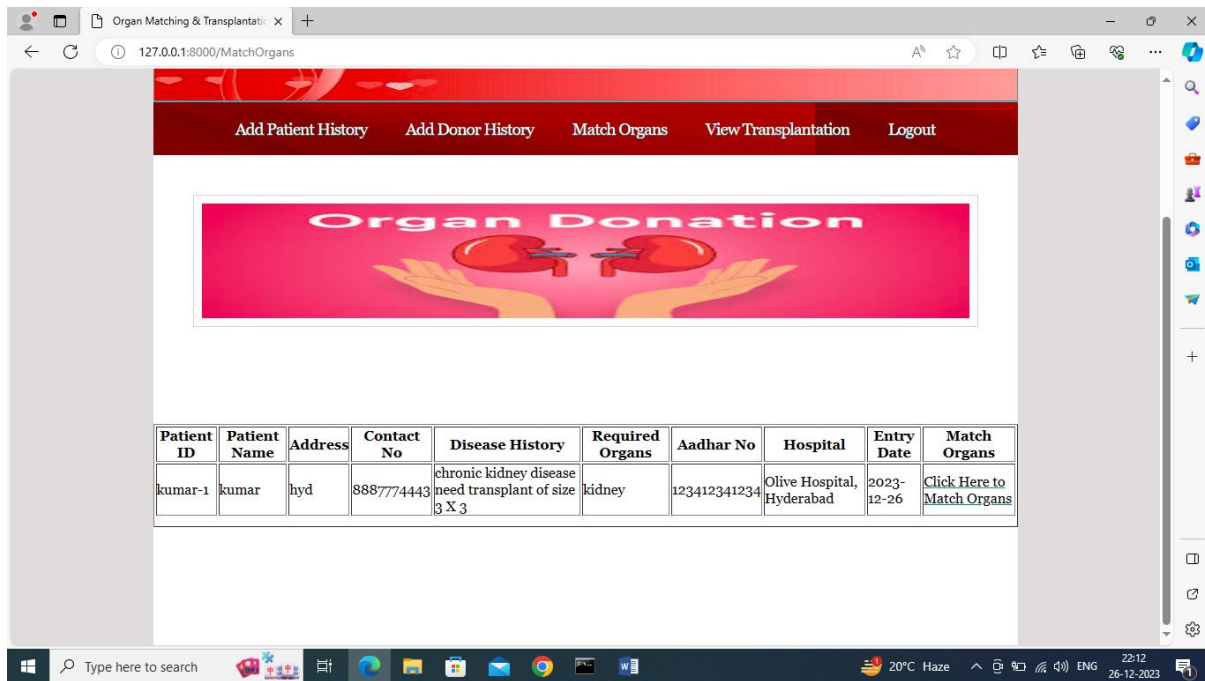


Fig 28: In above screen hospital can see all patients who need organs and can click on 'Click Here to Match Organs' link to view list of donors and then select desired donor

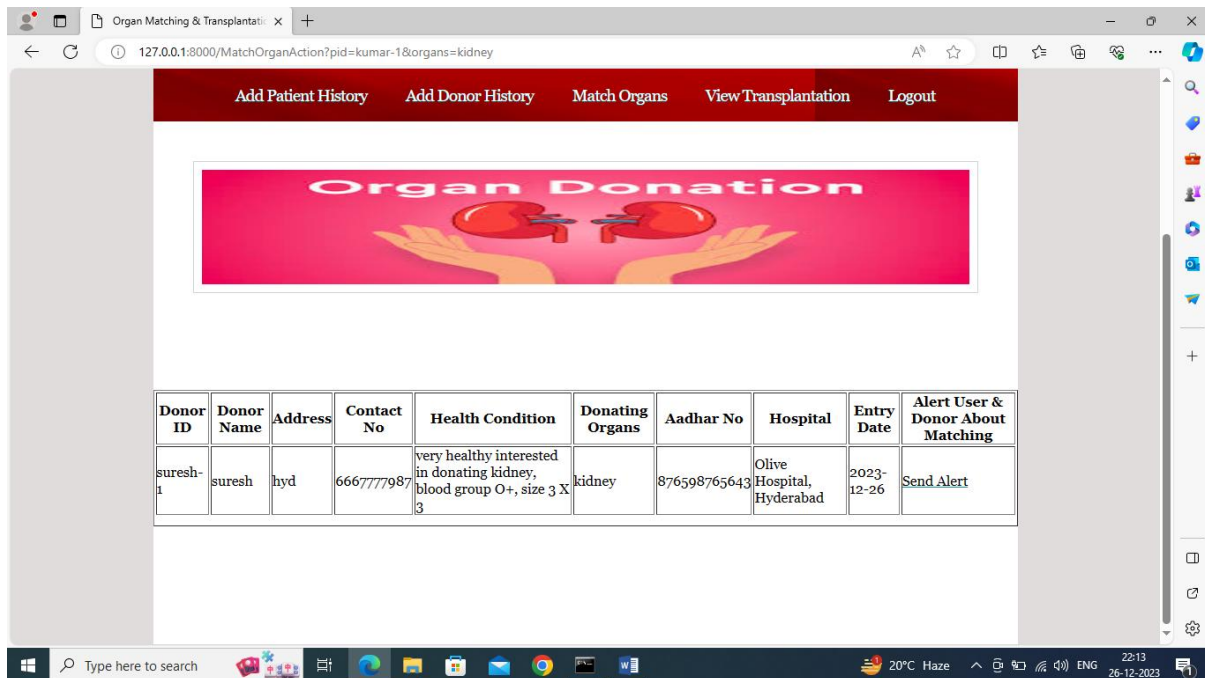


Fig 29: In above screen hospital can see one match and can click on 'Send Alert' link to send alert to both users and donors about match and then will get below output

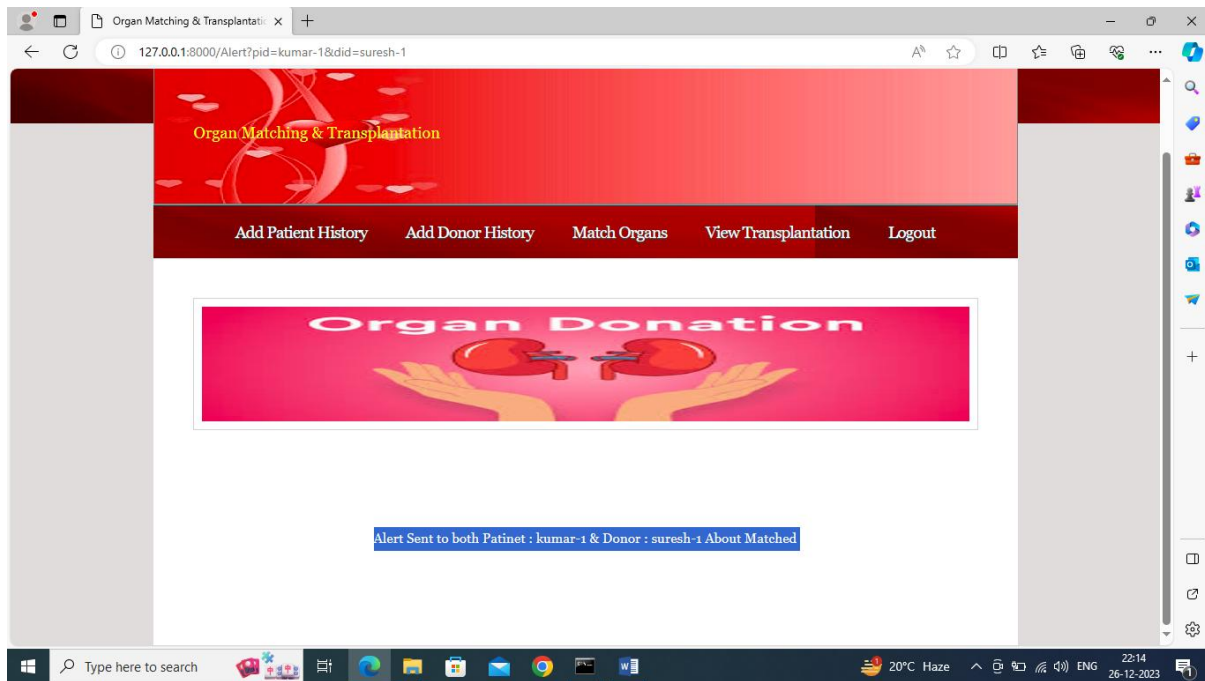


Fig 30: In above screen in blue colour text can see alert sent to both patients and donor and now hospital can click on 'View Transplantation' link to view all previous matches like below screen

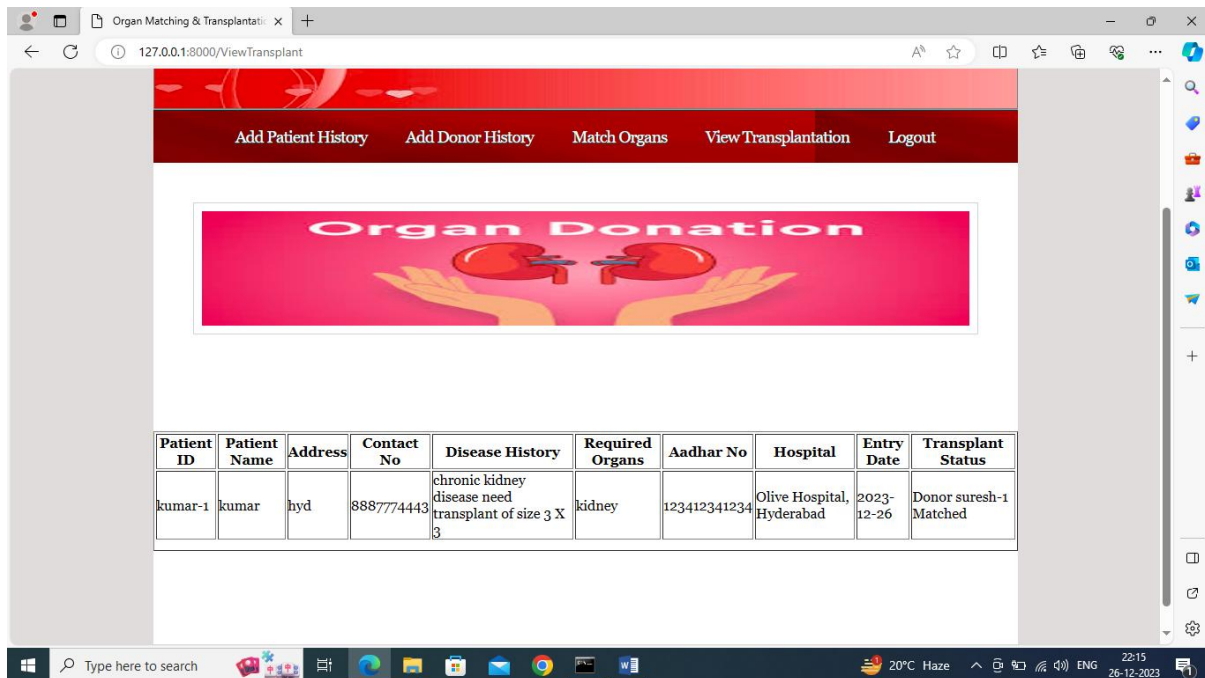


Fig 31: In above screen in last status column hospital can see which donor organ assigned to this patient. Now logout and login as donor to view status

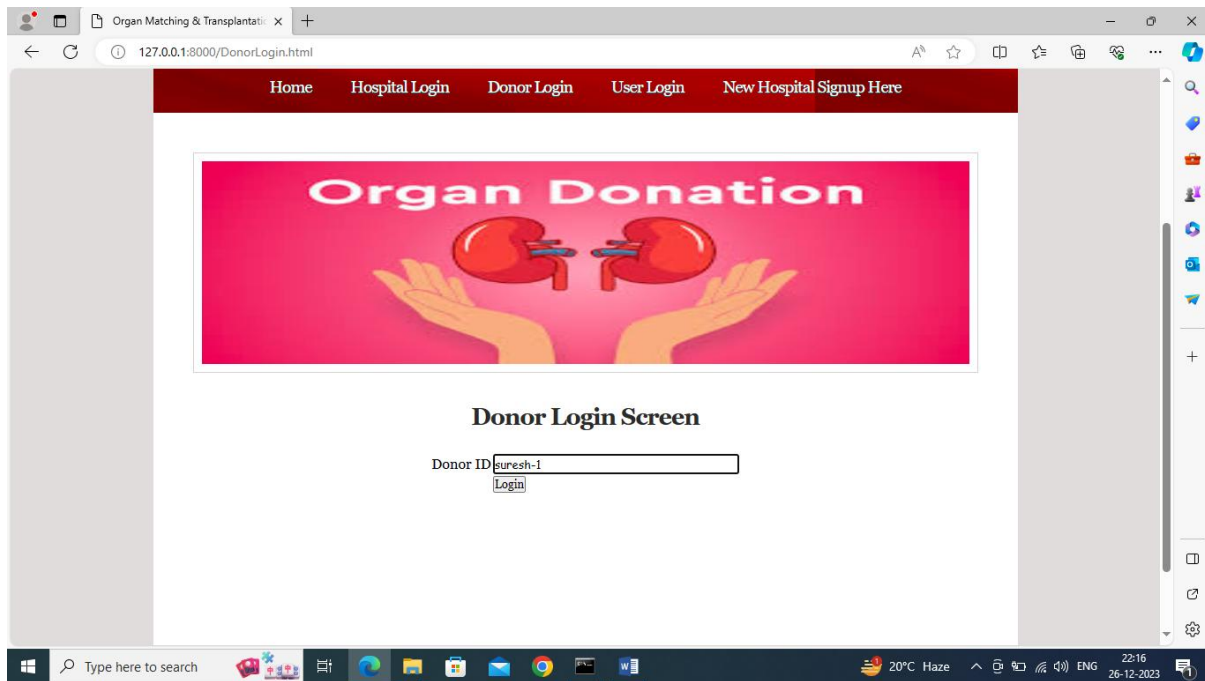


Fig 32: In above screen donor can enter his ID to get below page

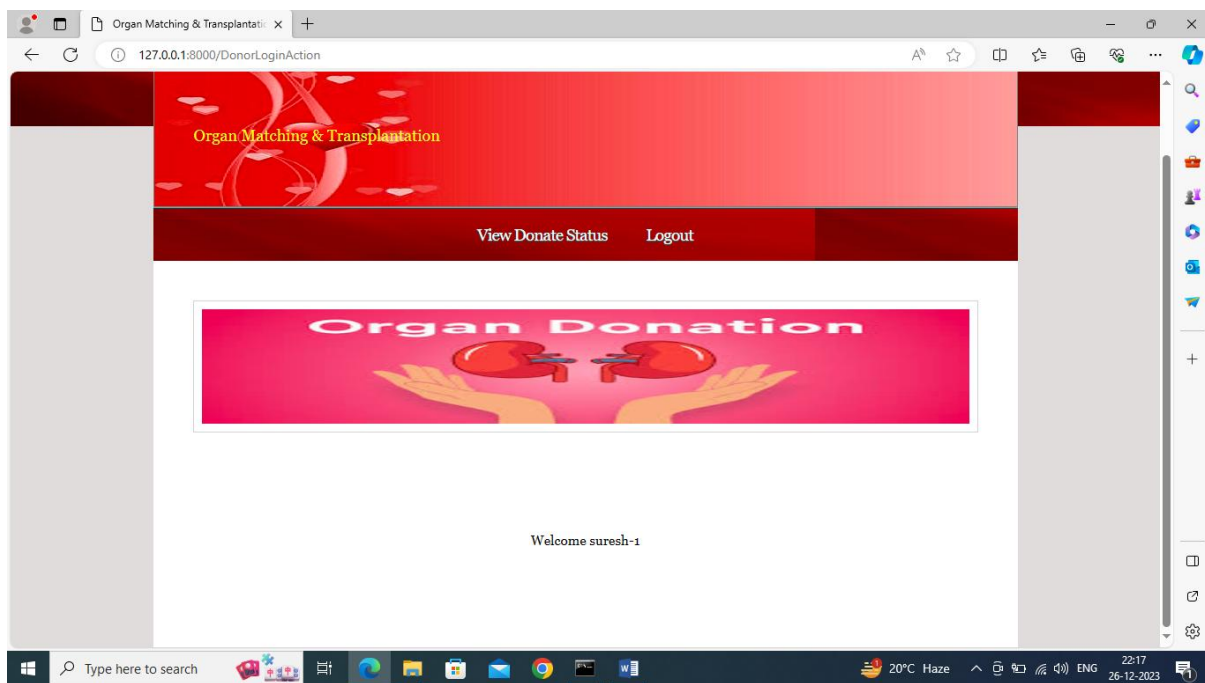


Fig 33: In above screen donor can click on 'View Status' link to get below organ matched status

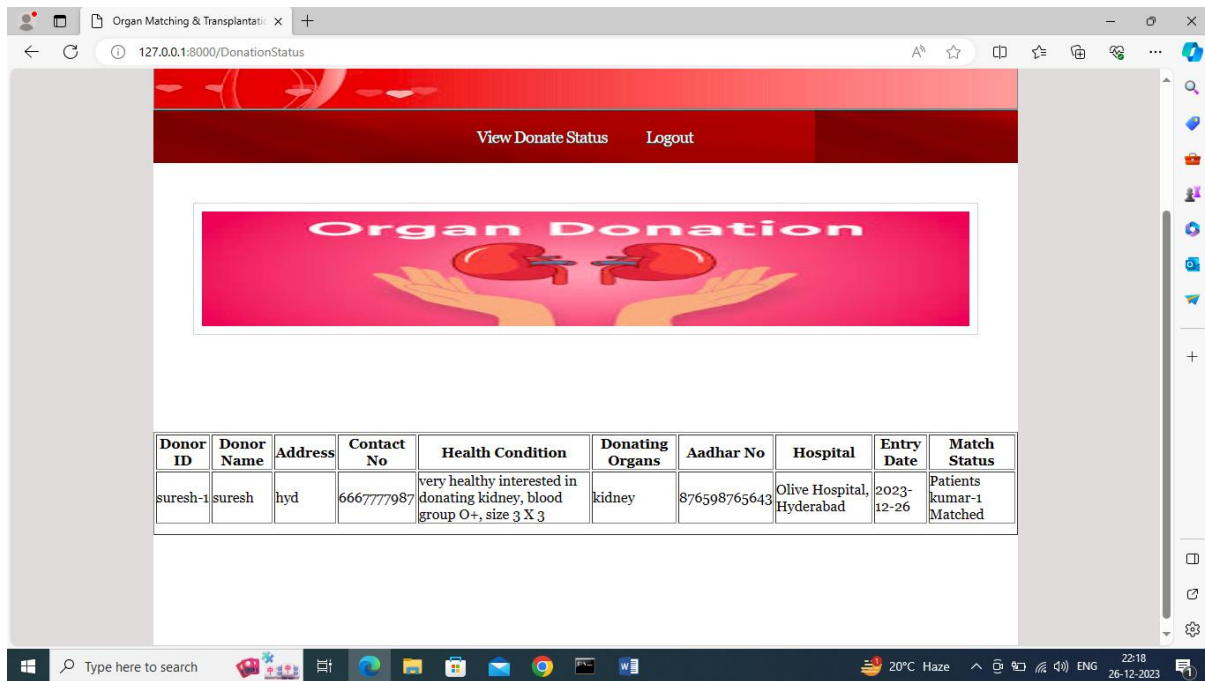


Fig 34: In above screen donor can see which patient is matched with his organ and now logout and login as ‘Patients or user’

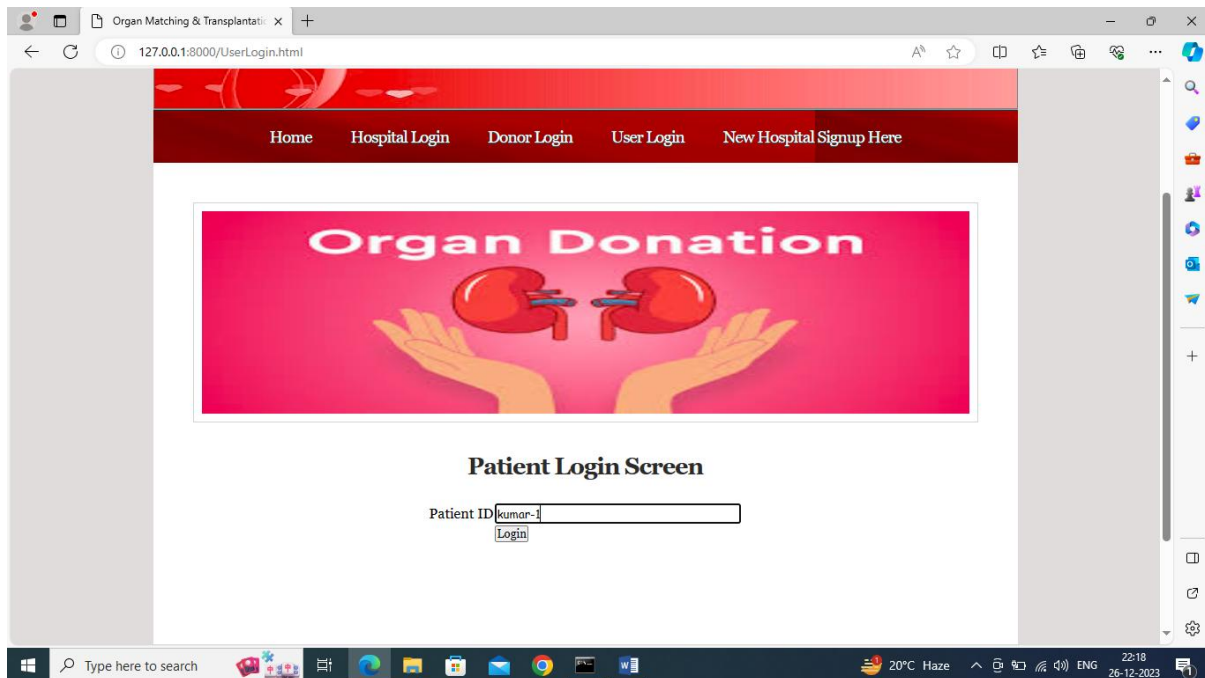


Fig 35: In above screen patient is login and after login will get below page

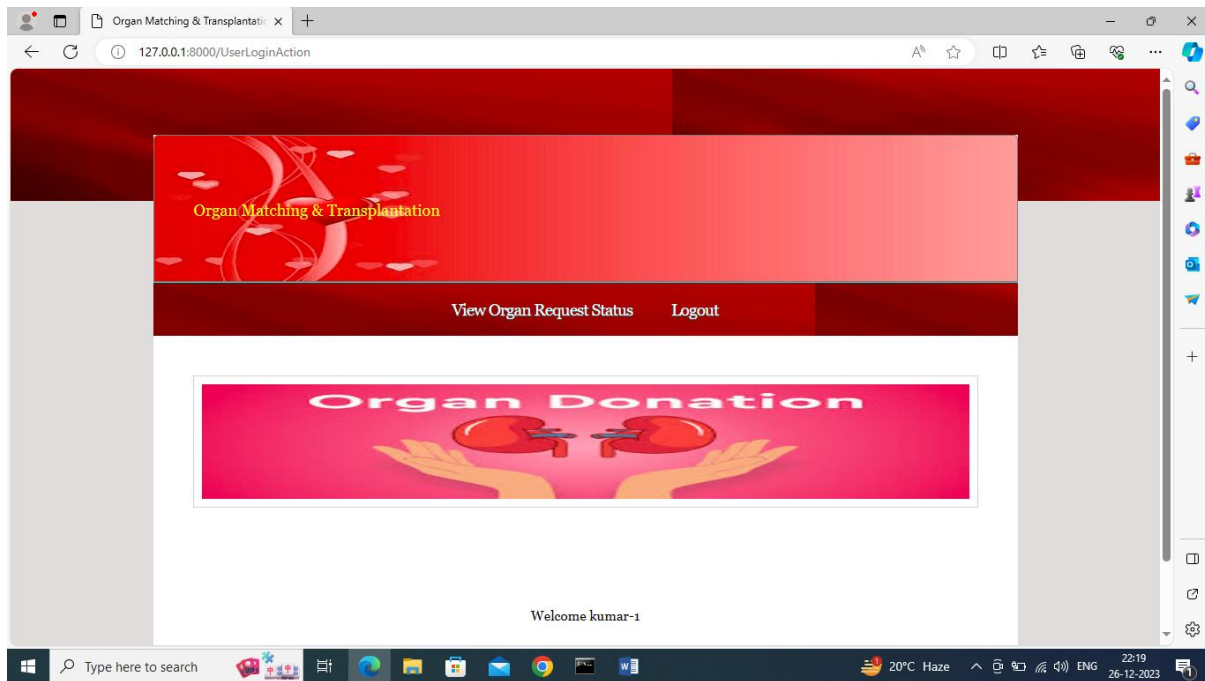


Fig 36: In above screen patient can click on 'View Organ Request Status' to view status of his organ request

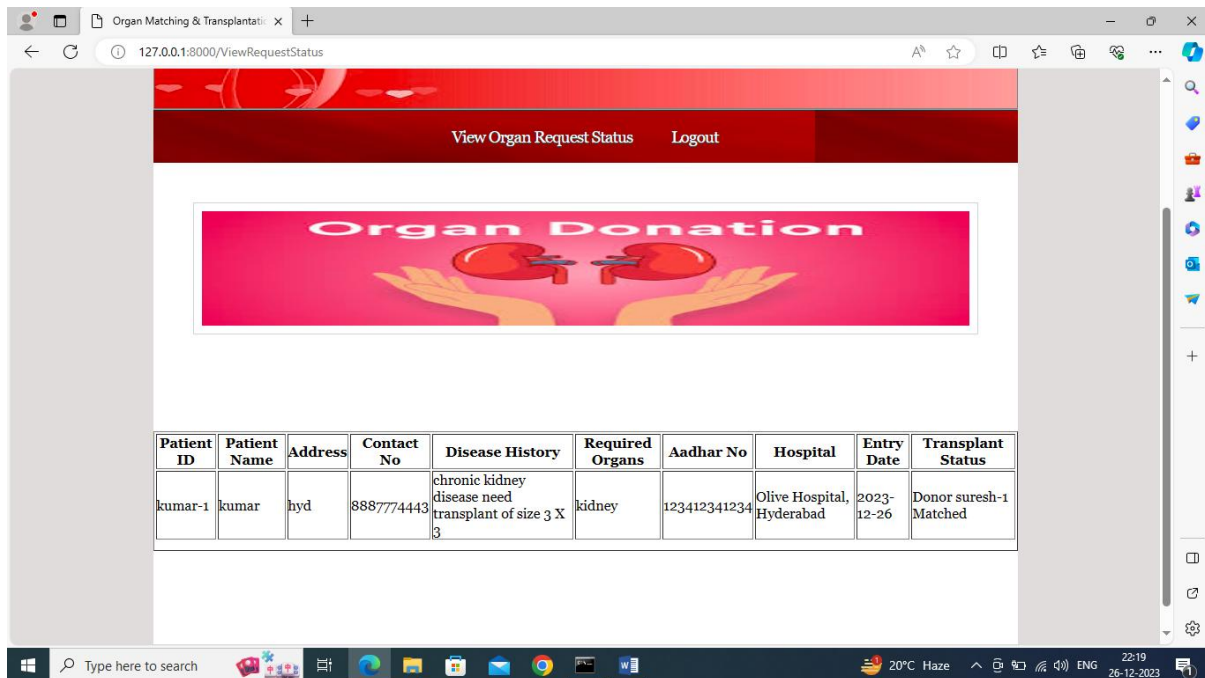


Fig 37: In above screen patient or user can view which donor organ matched with his request and all users and donors can view above alert once hospital matched them.
Similarly by following above screens you can save all details in Blockchain

CHAPTER 12

CONCLUSION

Conclusion:

In conclusion, the integration of blockchain technology into organ donation systems holds significant potential to revolutionize the way organs are allocated, tracked, and donated. By providing a secure, transparent, and immutable ledger, blockchain can enhance trust, reduce fraud, and ensure the equitable distribution of organs. The use of smart contracts can automate and streamline processes, increasing efficiency and reducing human error in organ matching and allocation. While the technical, economic, and social challenges to implementation are notable, the long-term benefits—such as improved transparency, reduced costs, and enhanced fairness—make blockchain a promising solution to address many of the existing issues in organ donation systems. With the proper infrastructure, education, and regulatory frameworks, blockchain has the potential to significantly improve the efficiency, equity, and trustworthiness of organ donation, ultimately saving lives and fostering a more transparent healthcare system. The system also enhances data security and privacy through encryption, preventing unauthorized access and minimizing the risk of illegal organ trade. With real-time updates and traceability, every step of the organ donation process can be monitored effectively. In conclusion, the implementation of blockchain technology in organ donation has the potential to revolutionize the healthcare sector by improving efficiency, ensuring transparency, and saving more lives. This project demonstrates how advanced technology can be used to address real-world problems and create a reliable and trustworthy system for the betterment of society.

CHAPTER 13

FUTURE ENHANCEMENT

The future scope of implementing blockchain in organ donation systems is vast, with the potential to transform the way organ donation and transplantation processes are managed globally. As blockchain technology continues to evolve, it could enable even more sophisticated features such as AI-driven organ matching, real-time monitoring of organ conditions during transportation, and the integration of biometric data for improved donor-recipient compatibility. The expansion of blockchain to international platforms could facilitate cross-border organ donations, improving access to life-saving transplants in regions with limited organ availability. Furthermore, advancements in blockchain interoperability could lead to seamless integration with global healthcare systems, creating a unified, transparent, and secure network for organ donation. As public awareness and trust in blockchain grow, the technology could become the standard for organ donation systems worldwide, ensuring faster, more equitable, and efficient allocation of organs, ultimately saving more lives.

- ☐ AI-based donor-recipient matching for better accuracy
- ☐ IoT integration for real-time health and organ monitoring
- ☐ Mobile application for easy access and updates
- ☐ Global blockchain network for international organ sharing
- ☐ Biometric authentication for enhanced security

CHAPTER 14

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REFERENCES:

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